

you^{do}_{the}
MA+HS

OLNA Support 2



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Warren Beckwith
You do the Maths



Money

GST

1.
 - (a) Determine the final price of an item which is worth \$60 prior to GST being added. GST is 10% of the original price.

 - (b) A car has a GST of \$2 500 included in its final price. How much is the car worth prior to the GST being added?

 - (c) What is the final amount if \$200 is increased by 10%, then the new amount is decreased by 10%?

2. Complete this table below for the following purchases, given that GST is 10% of the Price of the taxable item.

In each case, the final total is the original price plus GST.

Price of Taxable Item	GST	Total
\$100	\$10	\$110
\$10		
	\$2	
\$22.50		
\$35		
	\$10	
\$54.20		
\$145		
	\$1.50	
\$230.55		
\$1 094		
		\$220
		\$60.50

Change

Complete this table below for the following purchases:

Cash Given	Sale Price	Change
\$50	\$24.50	
\$100	\$79.80	
\$10		\$2.45
	\$23.60	\$6.40
\$85		\$2.95
	\$36.95	\$13.05
\$2	\$1.65	
	\$28.95	\$11.05
\$20		\$7.15
\$70	\$58.20	
	\$144.35	\$15.65
\$200		\$39.15

Discount

Complete this table below for the following purchases:

Original Price	Percentage Discount	Discount	New Price
\$50	10%		
\$240	20%		
\$300		\$15	
\$80			\$72
	5%	\$25	
		\$20	\$30
\$688	50%		
\$350			\$280
\$600	2%		
\$360			\$342
	10%	\$8.50	
	25%	\$12.25	

Practice OLNA Questions

Question 1

Ivan's lunch cost \$5.50. If he gave a \$10 note, how much change would he receive?

- | | | | |
|----------|--------|----------|--------|
| A | \$5.50 | B | \$4.50 |
| C | \$5.00 | D | \$6.50 |

Question 2

Mary-Lou was buying some apples.

Store One offers 12 apples for \$ 9.60.

Store Two sells five apples for \$ 4.10.

Store Three sells apples at 84 c each.

Store Four sells 6 apples for \$ 5.10

Which store offers the best buy?

- | | | | |
|----------|-------------|----------|------------|
| A | Store One | B | Store Two |
| C | Store Three | D | Store Four |

Question 3

Philip handed a \$10 for two coffees and received the following the change shown below.
How much did the two coffees cost?



- | | | | |
|----------|---------|----------|---------|
| A | \$ 9.60 | B | \$ 8.40 |
| C | \$ 8.60 | D | \$ 9.40 |

Question 4

How much money is shown below?



A \$ 3.35

B \$ 5.25

C \$ 2.40

D \$ 4.35

Question 5

Seamus is given the pocket money shown below, every day. How much has he been given after 110 days?



A \$ 390.50

B \$ 265.00

C \$ 291.50

D \$ 286.00

Question 6

Which of the following amounts offers the highest rate of interest?

A 10.2 % per annum

B 1.5 % per month

C 0.02 % per day

D 0.16 % per week

Question 7

A car was selling for \$ 12 800. A dealer then decided to have a 25 % discount sale. What is the new price for the car?

- | | | | |
|----------|----------|----------|-----------|
| A | \$ 6 400 | B | \$ 12 775 |
| C | \$ 8 500 | D | \$ 9 600 |

Question 8

Brent was charging a customer for his plumbing work. He charged \$ 65 for driving to the property, then \$ 56 per hour for 2 hours work. He then charged \$ 30 for parts. What was the total charge if 10% G.S.T was added to the final bill.

- | | | | |
|----------|-----------|----------|-----------|
| A | \$ 194.70 | B | \$ 156.20 |
| C | \$ 166.10 | D | \$ 227.70 |

Question 9

Sari made three purchases at her local shopping centre. She spent \$2.50 on a loaf of bread, \$36.85 on groceries and \$24.00 at the Butchers. How much change did she get from \$100?

- | | | | |
|----------|----------|----------|----------|
| A | \$ 36.65 | B | \$ 37.35 |
| C | \$ 46.65 | D | \$ 47.65 |

Michelle purchased three items at the Super Savers grocery store. The price for the tomatoes was printed out correctly, but an error caused the prices for the apples and capsicums to be left off the docket. Use the information below to answer Questions 10 to 11.

Super Savers	
JUICY APPLES	*. **
0.6 kg @ \$2/kg	
RED CAPSICUMS	*. **
0.4 kg @ \$6/kg	
FIELD GROWN TOMATOES	3.50
0.5 kg @ \$7/kg	

Question 10

How much did Michelle pay for the apples?

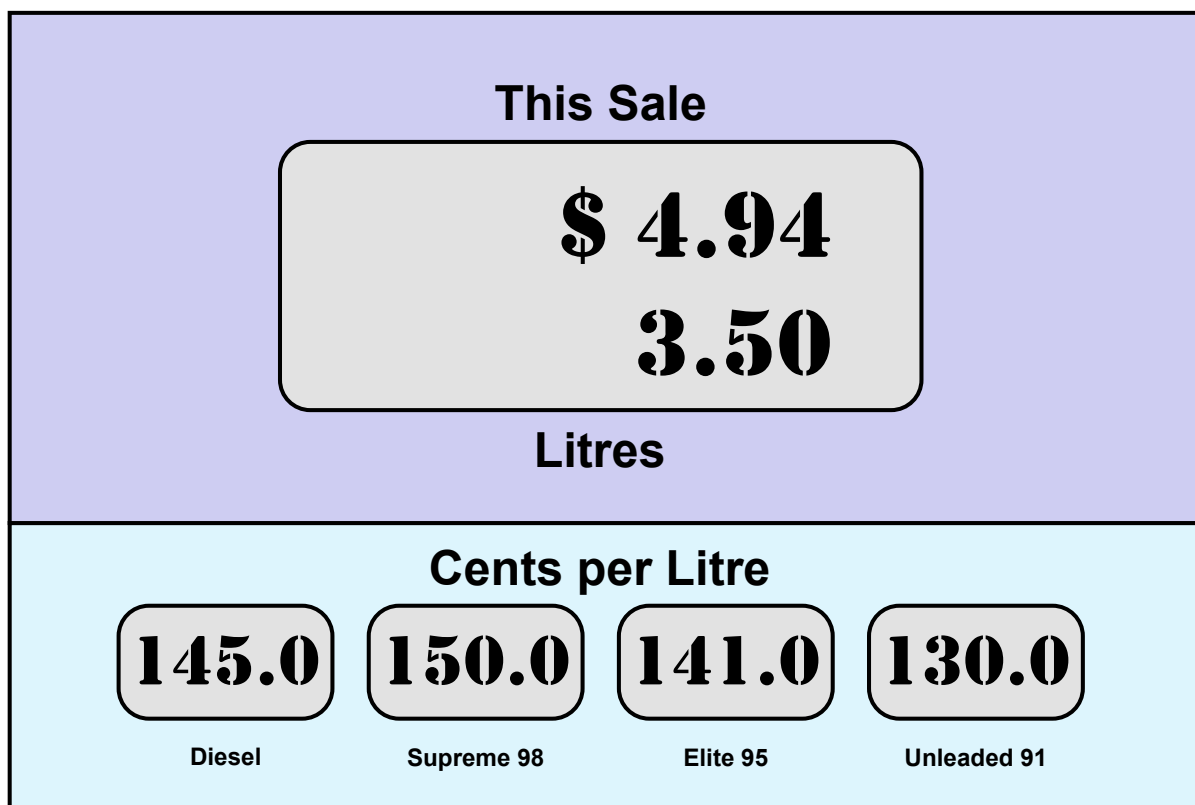
- | | | | |
|----------|--------|----------|----------|
| A | \$12 | B | 90 cents |
| C | \$1.20 | D | 60 cents |

Question 11

How much did Michelle pay for the capsicums?

- | | | | |
|----------|--------|----------|----------|
| A | \$6 | B | 80 cents |
| C | \$1.20 | D | \$2.40 |

Murali went to buy some fuel at his local petrol station. Consider the Petrol Bowser shown below to answer questions 12 to 15.



Question 12

Given the sales amount, and the number of litres in the diagram shown above, which type of fuel did Murali use?

- | | | | |
|----------|----------|----------|-------------|
| A | Diesel | B | Supreme 98 |
| C | Elite 95 | D | Unleaded 91 |

Question 13

Murali took another vehicle to the petrol station and purchased 20 litres of the Supreme 98 fuel. How much change did he get from \$ 50?

- | | | | |
|----------|------|----------|------|
| A | \$35 | B | \$20 |
| C | \$30 | D | \$15 |

Question 14

Whilst at the checkout, Murali noticed another driver use a fuel discount voucher whilst purchasing 50 litres of the Unleaded 91 fuel. If the voucher entitled them to a discount of 4 cents per litre, how much did they pay for their fuel?

- | | | | |
|----------|---------|----------|---------|
| A | \$73.00 | B | \$6.50 |
| C | \$65.00 | D | \$63.00 |

Question 15

On another visit Murali's petrol bowser was working correctly, but the digits showing the number of litres purchased was malfunctioning. If he purchased the diesel fuel for \$58, how many litres did he buy?

- | | | | |
|----------|----|----------|----|
| A | 14 | B | 20 |
| C | 35 | D | 40 |

Amy, Siti, Chris, Toshi and Hazam all went to the Café by the Sea for dessert after their main meal. The docket below shows their final bill. Use this to answer questions 16 to 18.

CAFE BY THE SEA	
Coffee - Flat White	4.00
Coffee - Black	4.00
Tea - Earl Grey	4.50
Tea - Peppermint	5.00
Hot Chocolate	4.50
Cake - Black Forest	5.50
Scones - Jam & Cream	5.00
GST	
TOTAL	

Question 16

Determine how much GST must be paid if it is 10% of the total of all of the items displayed on the docket.

- | | | | |
|----------|----------|----------|----------|
| A | \$ 32.50 | B | \$ 31.00 |
| C | \$ 3.25 | D | \$ 3.10 |

Question 17

Toshi ordered the flat white and the cake, whilst Hazam ordered the hot chocolate and scones. They each intended to pay for their meals separately. Amy, Siti and Chris decided to share the remainder of the bill equally. How much did Amy, Siti and Chris each pay, prior to GST being added?

A \$ 3.50

B \$ 4.00

C \$ 4.33

D \$ 4.50

Question 18

Chris was disappointed that his coffee was served cold. The manager of the café decided to deduct the GST from the bill and give Chris a free black coffee and a black forest cake at half price. (Toshi still had her black forest cake at full price.). What was the total of the bill after these adjustments were made?

A \$ 34.00

B \$ 29.00

C \$ 31.25

D \$ 35.25



Time

Calendars

1. Consider the calendar below for May, 2017.

Su	M	Tu	W	Th	F	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

- (a) Which days occur five times in the month?
 - (b) What fraction of the month is on a Thursday?
 - (c) If Stefan works from Monday to Friday, how many days does he work in this month?
 - (d) On what day of the week is June 8th, 2017?
 - (e) On what day of the week is April 27th, 2017?
2. Complete the calendar below, by placing dates in the vacant boxes, if..
- (i) the month has thirty-one days,
 - (ii) Thursday and Friday occur five times in this month,
 - (iii) the date on the first Saturday is one quarter of the date on the third Thursday.

Su	M	Tu	W	Th	F	Sa

Differences in Time

In each case below determine the difference in time, in minutes. Time 2 is the later time.

SET ONE

Time 1	Time 2	Difference (Minutes)
7:33 am	7:56 am	23 min.
12:11 pm	12:54 pm	
1:05 am	1:27 am	
5:01 am	5:39 am	
6:11 pm		25 min.
	11:56 pm	41 min.
10:16 am		33 min.
7:45 pm	7:59 pm	
	3:28 am	27 min.
	6:00 pm	34 min.

SET TWO

Time 1	Time 2	Difference (Minutes)
3:27 am	4:19 am	
5:16 am	6:08 am	
10:24 pm	11:36 pm	
1:19 am		50 min.
	4:23 pm	32 min.
12:16 am	1:23 am	
	4:01 pm	37 min.
	7:27 pm	78 min.
8:18 am		93 min.
	5:06 pm	64 min.

SET THREE

Time 1	Time 2	Difference (Minutes)
12:33 am	2:56 am	
12:24 pm	1:37 pm	
	2:44 pm	120 min.
12:39 pm		77 min.
12:04 am	1:01 am	
12:29 pm		125 min.
	3:19 am	181 min.
12:44 am	2:12 am	
	4:22 pm	255 min.
12:51 am	3:25 am	

SET FOUR

Time 1	Time 2	Difference (Minutes)
10:14 pm	1:20 am	
11:26 pm	12:35 am	
	2:19 pm	184 min.
	1:48 pm	98 min.
11:27 am		100 min.
9:38 am	1:15 pm	
	2:38 pm	189 min.
10:56 pm		85 min.
8:44 am		310 min.
6:23 pm	1:04 am	

Practice OLNA Questions

Use the calendar below to answer the questions 1 to 3.

February 2015

Su	M	Tu	W	Th	F	Sa
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

Sunday, the 1st, to Wednesday, the 4th, represents $\frac{4}{7}$ of the first week.

Question 1

Sunday, the 8th, to Monday, the 9th, represents what fraction of the second week?

A $\frac{4}{7}$

B $\frac{2}{7}$

C $\frac{2}{28}$

D $\frac{2}{5}$

Question 2

Friday, the 13th, to Monday, the 16th, represents what fraction of the month?

A $\frac{4}{7}$

B $\frac{5}{28}$

C $\frac{13}{28}$

D $\frac{1}{7}$

Question 3

What is the date a fortnight before the second Thursday in February, 2015?

A 29/1/15

B 5/2/15

C 26/2/15

D 28/1/15

Jeff has his birthday in March, use the calendar shown to answer the questions 4 to 5.

March 2016

Su	M	Tu	W	Th	F	Sa
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

Question 4

What is the probability that Jeff's birthday is an even day on the calendar?

A $\frac{1}{2}$

B $\frac{15}{31}$

C $\frac{16}{31}$

D $\frac{30}{31}$

Question 5

What is the probability that Jeff's birthday is on a day which is a multiple of 5?

A $\frac{5}{31}$

B $\frac{1}{5}$

C $\frac{30}{31}$

D $\frac{6}{31}$

Kate is about to take off on a flight and observes the list of departures shown below. Use this table to answer questions 6 to 8.

Destination	Airline	Flight	Time	Status	Gate
Broome	OzAir	OA244	9:34 am	On Time	A2
Adelaide	JetStream	JS1652	9:50 am	On Time	G1
Sydney	JetStream	JS1433	10:00 am	Delayed	G6
Port Hedland	OzAir	OA291	10:05 am	On Time	A1
Melbourne	JetStream	JS2908	10:25 am	On Time	G3
Canberra	JetStream	JS1361	10:35 am	On Time	G5
Darwin	OzAir	OA301	10:50 am	Delayed	A2

Question 6

If Kate reads the departures board at 8:49 am, how much time is there before the scheduled flight to Broome departs?

- | | |
|---------------------|---------------------|
| A 43 minutes | B 85 minutes |
| C 37 minutes | D 45 minutes |

Question 7

If the plane to Darwin is delayed for a further 84 minutes, what will be the new departure time?

- | | |
|-------------------|-------------------|
| A 11:34 am | B 12:14 pm |
| C 11:54 am | D 12:24 pm |

Question 8

OzAir flies to Broome every 150 minutes. When is the last flight on this day, given that it does not fly later than 8:00 pm?

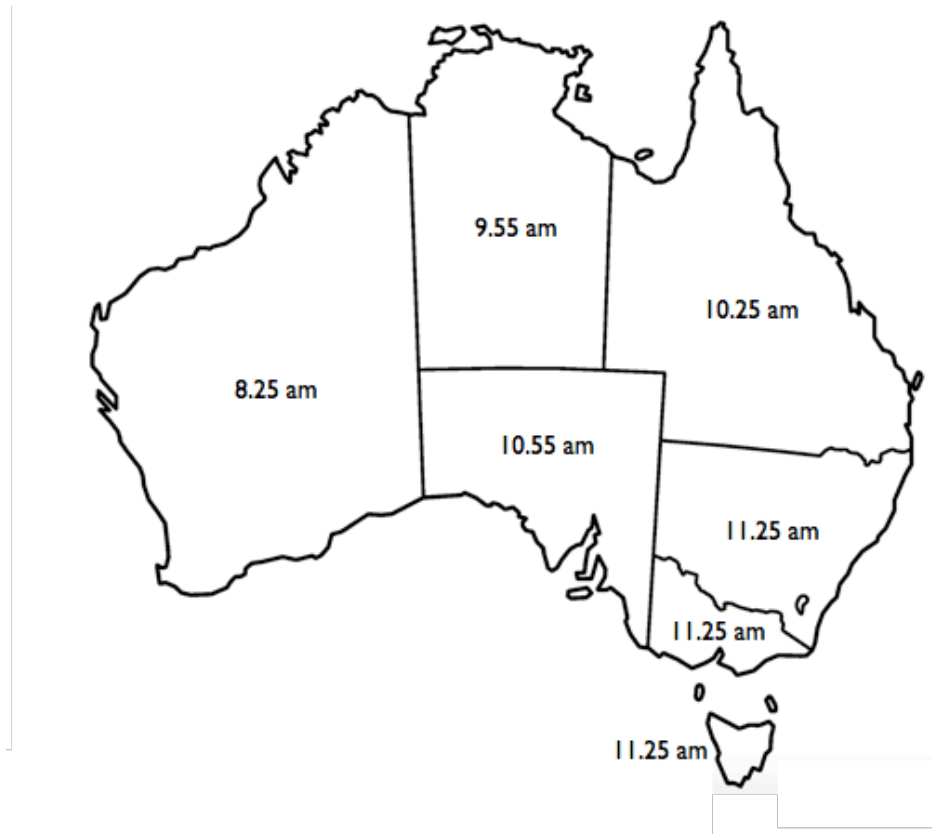
- | | |
|------------------|------------------|
| A 6:34 pm | B 6:04 pm |
| C 7:34 pm | D 7:04 pm |

Question 9

Three days ago, Sandy said to Mary, "I will call you eight days from now." Tomorrow is Sunday. What day will Sandy call?

- | | |
|-------------------|--------------------|
| A Tuesday | B Wednesday |
| C Thursday | D Friday |

Samantha runs a business which has branches in each state and territory of Australia. She needs to be aware of the different time zones when she co-ordinates meetings for each of the state directors. Susan is based in New South Wales. The map below shows the different times on a summer's day when it is 8.25 am in Perth, Western Australia. Use this information to answer questions 10 to 12.



Question 10

Samantha sat down to have her lunch at noon when someone called to say they have just finished their 8.30 am thirty minute meeting. Where was the caller coming from?

- | | |
|----------------------------|-----------------------------|
| A South Australia | B Queensland |
| C Western Australia | D Northern Territory |

Question 11

Samantha took a plane flight to Darwin, in the Northern Territory. If she left Sydney at 8.45 am and the flight took three and a half hours, what was the local time when she landed?

- | | |
|-------------------|-------------------|
| A 9.15 am | B 9.45 am |
| C 10.45 am | D 12.15 pm |

Question 12

Samantha decided to have an afternoon meeting for all of the state directors. She told them to be prepared for the meeting to last 90 minutes. Samantha was in Sydney when she started the meeting at 3.15 pm. At what time did the meeting finish for Jack, who is based in Perth?

A 12.15 pm

B 1.05 pm

C 1.45 pm

D 7.45 pm

Question 13

How many minutes have passed between the times shown on these two clocks, if both of these times occurred in the afternoon on the same day?



A 315

B 30

C 250

D 195

Question 14

Ashley is about to leave home for a train and looks at the clock in her kitchen. She knows that trains leave her local station at two minutes past the hour and every fifteen minutes after that. If she still has to put her shoes on (which takes two minutes), take out the rubbish (three minutes) and walk to the station (thirteen minutes), state the time she will leave her station on a train.



A 10:47 am

B 11:02 am

C 11:17 am

D 11.32 am

Question 15

Mark leaves his home, at a time given by the clock on the left. He travels by taxi to the airport, plane to Broome and then another taxi to his hotel. The next morning he checks the clock in the hotel, shown on the right, before heading down to Cable Beach. How many minutes have passed since he left home?



A 633

B 573

C 933

D 498

Question 16

Sophie has a coffee voucher that will expire in 24 days. If she notices this on July 18, on what day will her voucher expire?

- | | | | |
|----------|-----------|----------|-----------|
| A | August 12 | B | August 14 |
| C | August 10 | D | August 11 |

Question 17

Match each date below with its correct position in time, where the first date is the earliest and the fourth date is the latest.

Dec 31, 1998

Nov 29, 1998

Oct 14, 1999

Jan 4, 1999

- | | | | |
|----------|-------|----------|--------|
| A | first | B | second |
| C | third | D | fourth |

Question 18

What is the time, eighty-three minutes before 12.12 am?

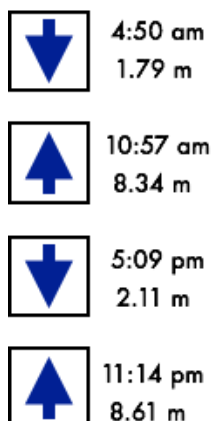
- | | | | |
|----------|----------|----------|----------|
| A | 1.35 am | B | 11.29 pm |
| C | 10.35 pm | D | 10.49 pm |

Question 19

If the month after the next month is June, what was the month before the previous month?

- | | | | |
|----------|----------|----------|-------|
| A | February | B | March |
| C | April | D | May |

The diagram below shows the times and measurements for low and high tides on one day at the Sunny Bay Beach Resort.



Question 20

What is the difference between when the first low tide occurs to when the first high tide occurs on this day?

- | | | | |
|----------|-------------|----------|-------------|
| A | 365 minutes | B | 367 minutes |
| C | 739 minutes | D | 737 minutes |

Question 21

Kristy leaves for a holiday on the second Thursday in February. She comes back a fortnight later on what date?

February 2017

Su	M	Tu	W	Th	F	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28				

- | | | | |
|----------|------|----------|------|
| A | 9th | B | 16th |
| C | 23rd | D | 24th |

Question 22

The current time is shown on the clock below. What was the time forty-five minutes ago?



A 5:40

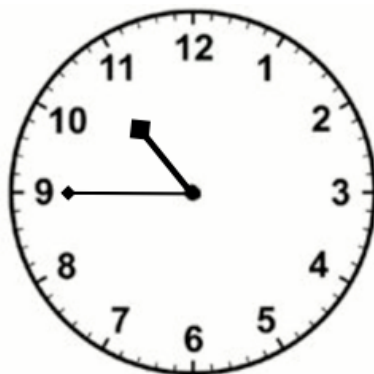
B 3:48

C 6:40

D 4:48

Question 23

The clock below shows the time that a bus departs Midland Bus Terminal. If the bus takes a thirty-eight minute journey, when would it arrive at it's destination?



A 11:13

B 11:23

C 11:33

D 11:18



Number

Quick Drills

7	X	11	=
5	X	8	=
9	X	3	=
2	X	12	=
6	X	12	=
11	X	10	=
9	X	4	=
3	X	3	=
2	X	4	=
7	X	3	=

3	X	4	=
9	X	5	=
4	X	6	=
12	X	6	=
2	X	12	=
6	X	12	=
9	X	11	=
12	X	2	=
9	X	6	=
10	X	12	=

20	÷	10	=
2	÷	1 000	=
30	÷	100	=
9 000	÷	1 000	=
9	÷	100	=
7	÷	10 000	=
800	÷	10	=
6	÷	10	=
0	÷	10	=
300	÷	1 000	=
300	÷	10 000	=
1	÷	100	=
1 000	÷	10 000	=
90	÷	1 000	=
400	÷	100	=
8	÷	1 000	=

8	÷	10	=
100	÷	10	=
600	÷	100	=
6	÷	10 000	=
5 000	÷	100	=
8 000	÷	10 000	=
500	÷	100	=
30	÷	100	=
80	÷	100	=
80	÷	10	=
4	÷	10	=
9 000	÷	1 000	=
60	÷	1 000	=
700	÷	1 000	=
2	÷	10 000	=
7	÷	1 000	=

200	÷	10	=	
400	÷	10	=	
8 000	÷	10 000	=	
6 000	÷	1 000	=	
7 000	÷	10 000	=	
0	÷	10 000	=	
200	÷	1 000	=	
9 000	÷	10	=	
600	÷	1 000	=	
0	÷	1 000	=	
40	÷	10 000	=	

6	÷	10	=	
300	÷	1 000	=	
0	÷	10	=	
7	÷	10	=	
400	÷	100	=	
100	÷	1 000	=	
6 000	÷	10	=	
10	÷	10	=	
500	÷	10	=	
800	÷	10 000	=	
7 000	÷	1 000	=	

27	+	47	=	
14	+	29	=	
3	+	19	=	
15	+	17	=	
26	+	36	=	
8	+	26	=	
23	+	16	=	
10	+	44	=	
28	+	11	=	
6	+	48	=	
13	+	36	=	
17	+	26	=	
20	+	29	=	
2	+	12	=	
21	+	23	=	

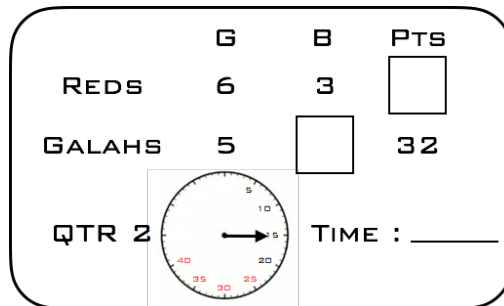
12	+	35	+	34	=	
14	+	27	+	26	=	
2	+	32	+	23	=	
16	+	27	+	31	=	
11	+	23	+	22	=	
6	+	39	+	19	=	
8	+	34	+	38	=	
8	+	32	+	21	=	
16	+	28	+	23	=	
1	+	21	+	26	=	
9	+	19	+	31	=	
8	+	39	+	28	=	
15	+	33	+	19	=	
3	+	25	+	19	=	
10	+	34	+	20	=	

32	-	5	=	
33	-	3	=	
25	-	13	=	
43	-	17	=	
45	-	22	=	
48	-	29	=	
32	-	15	=	
26	-	23	=	
20	-	6	=	
46	-	17	=	
20	-	4	=	
35	-	12	=	
37	-	18	=	
40	-	23	=	
42	-	13	=	

19	+	31	-	24	=	
13	+	33	-	19	=	
3	+	39	-	31	=	
31	-	24	+	28	=	
30	-	23	+	29	=	
15	+	34	-	33	=	
40	-	22	+	21	=	
44	-	29	-	9	=	
19	+	21	-	25	=	
50	-	31	+	27	=	
45	-	30	-	6	=	
41	+	33	-	30	=	
42	-	19	-	16	=	
20	+	37	-	35	=	
42	-	27	-	8	=	

AFL : Scoring

In Australian Football, scores are made up of goals (worth 6 points) and behinds with one point. Check the scoreboard below.

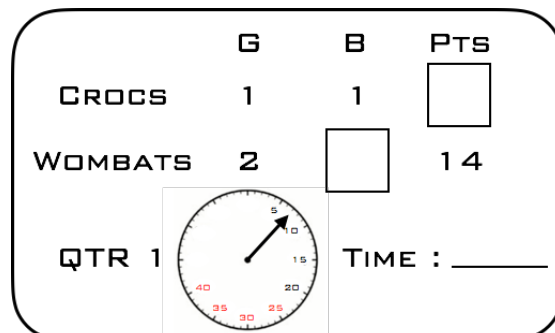
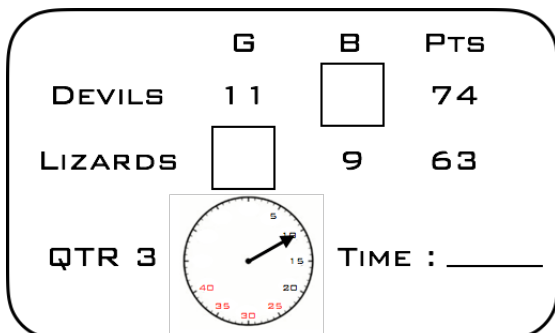
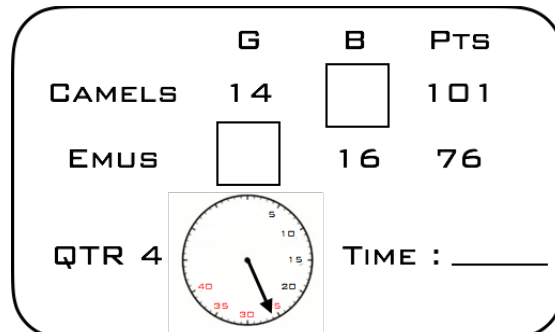
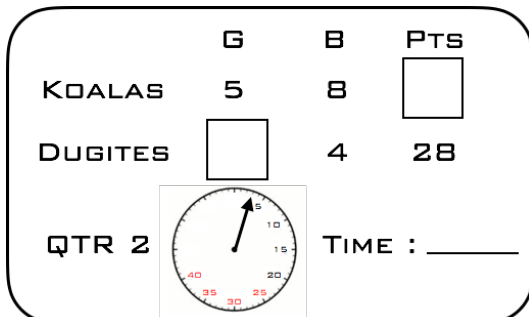


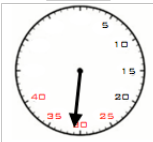
The scoreboard above shows the Reds on 6 goals, 3 behinds. This is a score of $6 \times 6 + 3$ or 39 points. The Galahs are on 32 points with 5 goals. 5 goals are worth 5×6 or 30 points. Therefore the Galahs scored 2 behinds.

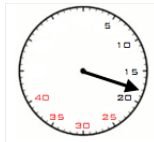
The scoreboard shows the time of the game.
In this case its 15 minutes (into the 2nd quarter.)

QUESTION ONE

Complete the following scoreboards by filling in the blank boxes and the time.



	G	B	PTS
QUOKKAS	3	5	
BULLS		7	31
QTR 1			TIME : _____

	G	B	PTS
GOANNAS	8		66
ECHIDNAS		11	77
QTR 3			TIME : _____

QUESTION TWO

For each score below write down 3 options for how a team could achieve that total. For example, 33 points can be achieved with 5 goals and 3 behinds.

(a) 57 points

(b) 98 points

(c) 124 points

(d) 167 points

Recipes

1. Consider the recipe below for Custard Tarts, which serves 4 people :

2 eggs
 2 tablespoons of sugar
 1 cup of milk
 4 drops of vanilla essence
 1 quantity of shortcrust pastry
 1 tablespoon of jam
 A sprinkle of nutmeg

Now determine the ingredients for the number of people shown in the table below:

Ingredients	2 people	10 people	1 person	7 people
eggs				
sugar				
milk				
vanilla essence				
shortcrust pastry				
jam				

2. Consider the recipe below for Peach Melba, which serves 4 people :

2 ripe peaches
 150 mL pureed raspberries
 4 scoops of ice cream
 2 tablespoons of sugar
 100 mL of water
 1 teaspoon of vanilla

Now determine the ingredients for the number of people shown in the table below:

Ingredients	2 people	3 people	14 people	9 people
peaches				
raspberries				
ice cream				
sugar				
water				
vanilla				

Triple Treats

Rearrange the numbers listed below and use Order of Operations to match the correct total shown below.

Example :

Numbers : 5 16 8 Total : 7

Solution : $16 \div 8 + 5$

Numbers...			Total	Solution...
5	1	3	15	
12	6	4	6	
7	10	6	11	
5	13	6	17	
5	3	4	23	
7	9	3	30	
2	4	10	5	
9	3	7	21	
4	7	2	42	
10	3	2	8	

Awesome Foursomes

Rearrange the numbers listed below and use Order of Operations to match the correct total shown below.

Example :

Numbers : 7 8 12 5 Total : 23

Solution : $(12 - 8) \times 7 - 5 = 23$

Numbers...				Total	Solution...
5	7	10	2	40	
3	8	12	4	27	
6	10	4	1	20	
16	4	3	9	7	
10	8	7	4	34	
1	5	6	12	7	
8	5	4	2	18	
15	10	3	2	30	
24	6	4	1	0	
12	5	6	4	14	

Practice OLNA Questions

Question 1

Which is the greatest amount?

- | | | | |
|----------|-----------------|----------|------------------|
| A | 1.6 % of 10 000 | B | 0.1 % of 100 000 |
| C | 3.2 % of 6 000 | D | 1 % of 12 000 |

It takes approximately twenty drops to make 1 mL, when using the bottle and eye dropper shown below. The bottle has a capacity of 25 mL.
Use this information to answer questions 2 to 4.



Question 2

How many drops would it take to empty the bottle?

- | | | | |
|----------|-----|----------|-------|
| A | 100 | B | 250 |
| C | 500 | D | 2 500 |

Question 3

If 150 drops were taken from the bottle, how many millilitres would be remaining in the bottle?

- | | | | |
|----------|--------|----------|---------|
| A | 7.5 mL | B | 10 mL |
| C | 15 mL | D | 17.5 mL |

Question 4

A new bottle of size 40 mL was produced. It has an improved dropper which can empty the bottle after 600 drops. How many drops does it take to make 1 mL?

- | | | | |
|----------|----|----------|----|
| A | 5 | B | 15 |
| C | 10 | D | 12 |

Question 5

One third of peaches in a box will be discarded due to poor quality. The other 72 are considered up to standard for sale in a supermarket. How many peaches are in the box?

- | | | | |
|----------|-----|----------|-----|
| A | 218 | B | 108 |
| C | 90 | D | 75 |

Question 6

Add these numbers together : 0.136, 0.004 and 0.07

- | | | | |
|----------|-------|----------|-------|
| A | 1.236 | B | 0.21 |
| C | 0.840 | D | 0.147 |

Question 7

Two consecutive even numbers add to 34. What is the largest of the two numbers?

- | | | | |
|----------|----|----------|----|
| A | 17 | B | 32 |
| C | 20 | D | 18 |

Question 8

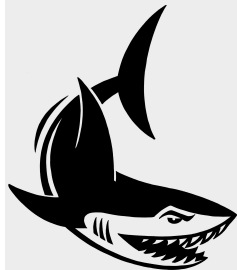
What numbers will be next in this sequence?

1, 5, 11, 19, 29, ...

- | | | | |
|----------|------------|----------|------------|
| A | 39, 49, 59 | B | 40, 52, 65 |
| C | 41, 53, 65 | D | 41, 55, 71 |

Kevin has developed an online game for Australian consumers. By taking pictures of the following animals, players can score points. The aim of the game is to get Australians to explore and understand the fauna of their own country.

Some of the animals are shown below. Use the scores shown to answer questions 9 to 11.

			
Kangaroo and joey	Koala	Tasmanian Devil	Great White Shark
30 Points	60 Points	200 Points	90 Points

Question 9

What score would Harry gain if he had two photos of koalas, one of a Tasmanian devil and one of a kangaroo with a joey

- | | |
|--------------|--------------|
| A 290 | B 380 |
| C 350 | D 320 |

Question 10

Larry was competing against Harry and decided to concentrate on sharks. He saw some sharks and was able to take two pictures of kangaroos with their joey. How many pictures of sharks did Larry take if his final score was 240.

- | | |
|------------|------------|
| A 0 | B 1 |
| C 2 | D 3 |

Question 11

Hilary decided to join in on the game. She had one picture of each animal with about half a day of playing time to go. She managed to take a picture of one more animal to attain a score of 440. Which animal did she manage to take a picture of at the end of the day?

- | | |
|------------------------------|----------------------------|
| A Kangaroo (and joey) | B Koala |
| C Tasmanian Devil | D Great White Shark |



Fractions and Percentages

Quick Drills

SET ONE

Determine the given percentage of each amount.

20 % of 200 =	40
50 % of 400 =	
10 % of 1 000 =	
90 % of 300 =	
5 % of 600 =	
60 % of 300 =	
5 % of 50 =	
35 % of 250 =	
85 % of 450 =	
80 % of 400 =	
65 % of 300 =	
75 % of 450 =	
55 % of 500 =	
10 % of 450 =	
80 % of 450 =	

SET TWO

Determine the amounts as a percentage.

40 out of 200 =	20%
5 out of 500 =	
25 out of 500 =	
89 out of 100 =	
240 out of 1 200 =	
30 out of 300 =	
45 out of 900 =	
75 out of 300 =	
225 out of 300 =	
200 out of 800 =	
10 out of 1 000 =	
2.5 out of 500 =	
40 out of 2 000 =	
5 out of 25 =	
16 out of 40 =	

Simplifying Fractions

Simplify the fractions below to complete the puzzle at the base of the page.

$$G \quad \frac{2}{10}$$

$$W \quad \frac{9}{15}$$

$$P \quad \frac{2}{18}$$

$$U = \frac{16}{24}$$

$$R = \frac{8}{20}$$

$$N \quad \frac{33}{45}$$

$$S \quad \frac{28}{44}$$

$$\begin{array}{r} 32 \\ Y \overline{) 24} \end{array}$$

What simplest fraction is...

L 14 of 30

20 c of \$2

D 300 mL of 1 Litre

0.5 of 10

H 3 kg of 1 200 g

A 5 cm of 1.2 m

T Forty-five minutes of half a day **E** A weekend of Autumn

Legend

Match the answer to its position in the order

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

$\frac{1}{46}$	$\frac{1}{5}$	$\frac{1}{20}$	$\frac{11}{15}$	$\frac{1}{9}$	$\frac{4}{3}$	$\frac{7}{11}$	$\frac{2}{5}$	$\frac{5}{2}$	$\frac{1}{24}$	$\frac{2}{3}$	$\frac{3}{5}$	$\frac{1}{10}$	$\frac{1}{16}$	$\frac{3}{10}$	$\frac{7}{15}$
----------------	---------------	----------------	-----------------	---------------	---------------	----------------	---------------	---------------	----------------	---------------	---------------	----------------	----------------	----------------	----------------

12	9	10	14		15	13	15		14	9	1		9	10	14		7	10	6		14	3		14	9	1	
14	1	16	1	5	9	3	4	1		6	3	11		7	14	10	6		9	1	8	1		13		16	16
									?																		
2	3		3	4		10	9	1	10	15																	

Fractions of Quantities

Solve the questions below to complete the puzzle at the base of the page.

H $\frac{1}{2}$ of 24

Y $\frac{1}{3}$ of 27

C $\frac{4}{5}$ of 35

M $\frac{4}{7}$ of 42

G $\frac{5}{6}$ of 36

E $\frac{1}{3}$ of 42

L $\frac{1}{3}$ of 2 hours (in minutes)

U $\frac{1}{4}$ of 2 m (in cm)

K $\frac{2}{18}$ of \$5.40 (in cents)

R $\frac{1}{25}$ of 2 kg (in g)

T $\frac{11}{12}$ of 96

N $\frac{12}{15}$ of 105

D $\frac{4}{7}$ of 56

W $\frac{9}{15}$ of 75

I $\frac{4}{20}$ of 360

A $\frac{3}{14}$ of 70

O $\frac{7}{26}$ of 78

45	12	15	88		32	21		9	21	50		28	15	40	40		88	45	21		24	14	84
40	21	21	60	72	84	30		21	50	88		15		45	72	84	32	21	45				
60	50	80	88		15	84	32		80	21	32												

Practice OLNA Questions

The cake tray shown below is to be used to cook cupcakes. Use this diagram to answer questions 1 to 3.



Question 1

If one quarter of the tray was filled with a sponge cake recipe, how many sponge cupcakes would you have?

A 4

B 8

C 7

D 6

Question 2

One and half dozen small cupcakes were made using a chocolate cake recipe. What fraction of the tray does this represent?

A $\frac{1}{2}$

B $\frac{1}{3}$

C $\frac{2}{3}$

D $\frac{3}{4}$

Question 3

Another batch of cupcakes was used using a tray of the same size. One third of the tray had cake with chocolate pieces in the recipe, 25% of the tray had cake with lemon in the recipe. Assuming no cakes had a mixture of chocolate or lemon, how many cakes did not include the chocolate pieces or lemon?

A 10

B 15

C 8

D 5

Questions 4 to 5 refer to the uploading progress bar shown below.



Question 4

Shaminda was waiting for a file to be uploaded on her laptop and estimated the percentage of the upload that had already occurred, based on the diagram above. State what you think would be the best estimate?

- | | | | |
|----------|------|----------|------|
| A | 35 % | B | 85 % |
| C | 15 % | D | 45 % |

Question 5

Later Shaminda noticed that 24 MB had been uploaded, and the uploading progress bar showed that 75% of the upload was complete. What was the total size of the file she was uploading?

- | | | | |
|----------|-------|----------|---------|
| A | 18 MB | B | 16.8 MB |
| C | 30 MB | D | 32 MB |

Question 6

The opposite faces on a dice add up to seven. What is the total sum of the numbers which cannot be seen on the two dice shown below.



A 12

B 26

C 16

D 42

Question 7

When two dice are rolled, what is the probability that a sum of eight will occur?

A $\frac{5}{36}$

B $\frac{1}{12}$

C $\frac{8}{12}$

D $\frac{8}{36}$

Question 8

Half of 60 is added to 30% of 300. What is the result?

A 60

B 90

C 40

D 120

Measurement and Conversions

Conversions

1. Complete the following tables:

Length (km)	Length (m)
6	
	900
3	
	100 000
500	
7	
	200
0.6	
	30 000
1	
0.4	
	8 000
0.7	
	8 100
	100

Length (cm)	Length (mm)
9	
	60
100	
	70
900	
50	
	7 000
8 000	
	7 000
8 020	
1	
	3 000
0.8	
	800
	300

2. WHICH IS LONGER?

- (i) 29 cm, 288 mm, 0.27 m
- (ii) 3.7 km, 36 000 cm, 38 000 000 mm
- (iii) 56 cm, 0.06 m, 526 mm

3. Add these amounts together and give your answer in cm.

- (i) 46 cm, 2.3 m, 1 km
- (ii) 200 mm, 46 cm, 6 m
- (iii) 3 km, 400 mm, 0.6 m, 38 cm
- (iv) 6 080 mm, 4.33 m, 86 cm
- (v) 5.76 km, 320 m, 46 000 mm

4. Complete the following tables:

Capacity (L)	Capacity (mL)
6	
4.2	
	8 000
300	
	65
0.03	
	40 000
	23 000 000
0.04	
	31 277

Time (hours)	Time (minutes)
4	
1.5	
	840
6 h 20 min	
	450
3.25	
0.2	
	135
	1 200
6.75	

Practice OLNA Questions

Question 1

Match the 1 Litre bottles with each measurement of milk shown below



A 700 mL

B $\frac{3}{4}$ L

C 10% of 1 Litre

D 0.5 L

Question 2

A rectangular paddock has a length of 350 m and a width of 1 km. What is the perimeter?

A 0.35 km

B 2.7 km

C 1.7 km

D 1 350 m

The diagram below shows the times and measurements for low and high tides on one day at the Sunny Bay Beach Resort. Use this information to answer question 3.

	4:50 am 1.79 m
	10:57 am 8.34 m
	5:09 pm 2.11 m
	11:14 pm 8.61 m

Question 3

What is the difference between the lowest tide and the highest tide on this day?

- | | |
|-----------------|-----------------|
| A 655 cm | B 623 cm |
| C 718 cm | D 682 cm |

Question 4

Rahul's height is 176 cm. Which height category does he fit into?

- | | |
|---|---|
| A Up to and equal to 160 cm | B More than 160 cm, up to 168 cm |
| C More than 168 cm, up to 175 cm | D More than 175 cm |

Question 5

What distance is the smallest in this group?

- | | |
|--------------------|------------------|
| A 3.8 m | B 0.36 km |
| C 10 700 mm | D 356 cm |

Question 6

A set of scales had some small weights. What is the combined total of these weights: 120 g, 0.3 kg and 4 200 mg?

- | | | | |
|----------|---------|----------|---------|
| A | 4 620 g | B | 462 g |
| C | 424.2 g | D | 454.2 g |

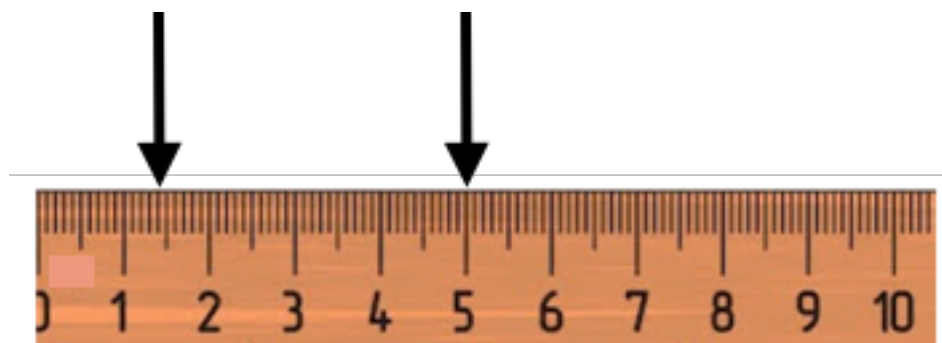
Question 7

When standing at full stretch, barefooted, Carlos can reach to a height of 203 cm. If his shoes extend his height by 17 mm and he is standing on a platform which is 0.4 m above the ground how far can Carlos reach at full stretch now?

- | | | | |
|----------|-----------|----------|----------|
| A | 2 447 mm | B | 2 087 mm |
| C | 20 717 mm | D | 6 047 mm |

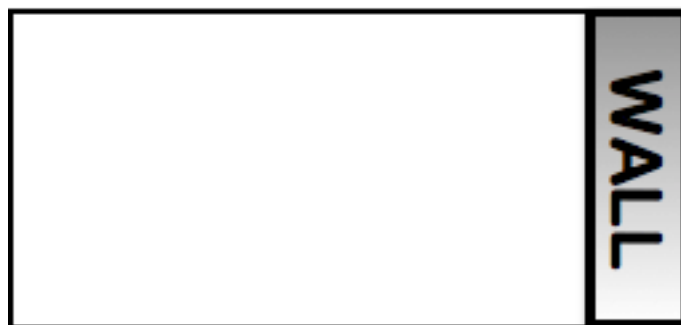
Question 8

The numbers shown on the ruler below represent centimetres. What is the approximate distance from one arrow to the next?



- | | | | |
|----------|-------|----------|-------|
| A | 44 mm | B | 46 mm |
| C | 34 mm | D | 36 mm |

Travis wanted to work out the perimeter of his rectangular yard, which is shown below. One side borders a wall, which Travis knows is 15 m long. Use this information to answer questions 9 to 10.



Question 9

Travis walked the entire side bordering the wall in 18 paces. He then walked alongside the adjacent longer side and did this in 27 paces. Assuming Travis maintains the same pace length. What is the perimeter of the yard, including the side with the wall?

- | | | | |
|----------|------|----------|------|
| A | 75 m | B | 90 m |
| C | 45 m | D | 60 m |

Question 10

Travis wants to put a walking path inside the yard. If the path is to be 25 m long, how many paces will it take Travis to walk the entire path?

- | | | | |
|----------|----|----------|----|
| A | 9 | B | 25 |
| C | 30 | D | 32 |

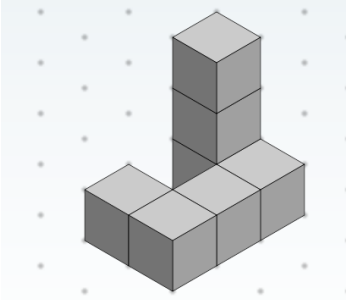
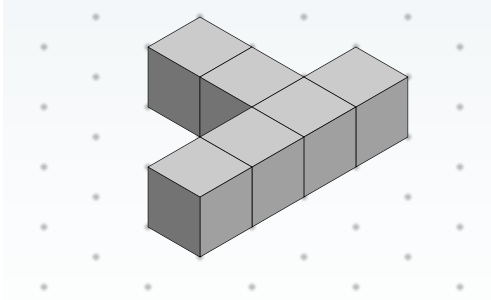
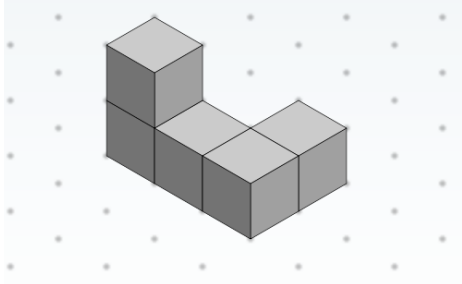
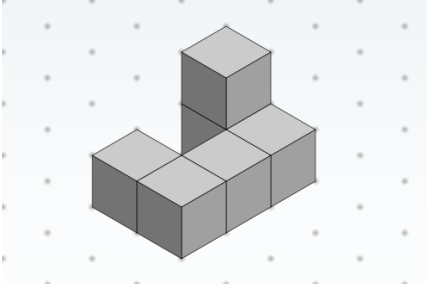
Question 11

Imogen used three 600 mL chocolate milk cartons to serve 7 people, who each had 250 mL of milk in cups. How much milk was left over?

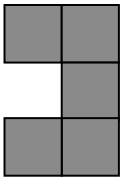
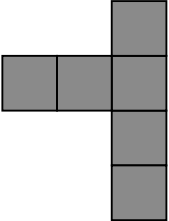
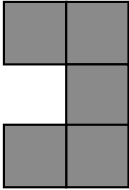
- | | | | |
|----------|--------|----------|--------|
| A | 50 mL | B | 150 mL |
| C | 100 mL | D | 200 mL |

Shapes

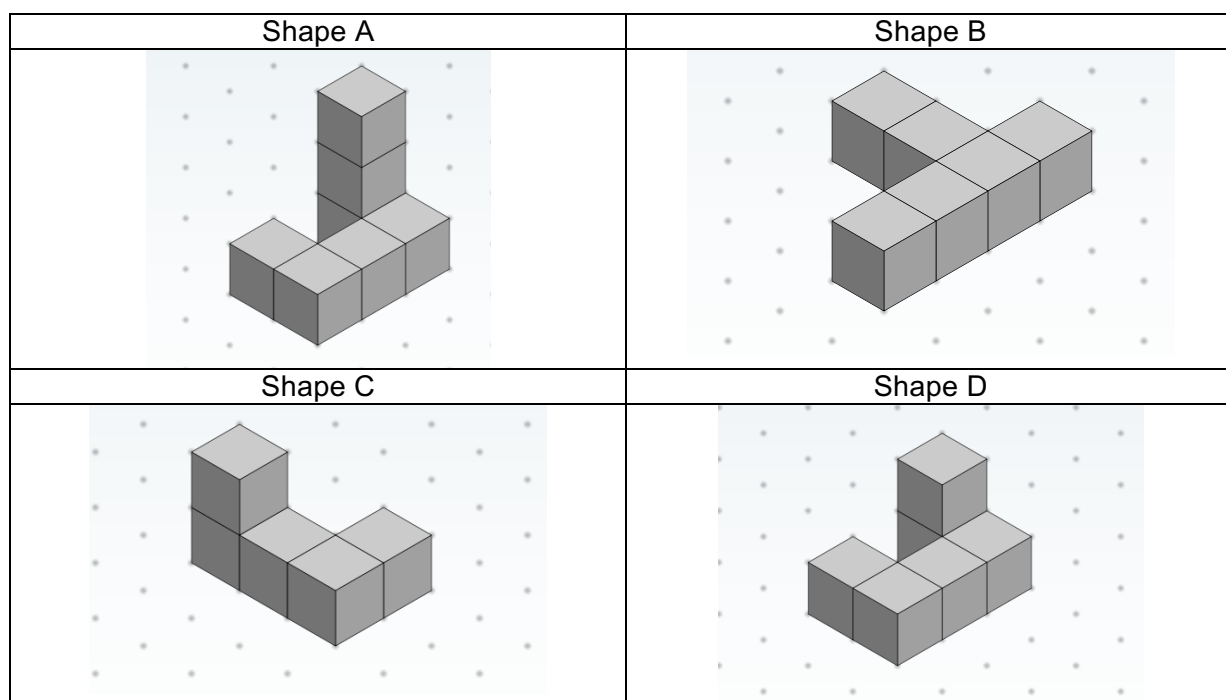
1. Consider each shape shown below.

Shape A	Shape B
	
Shape C	Shape D
	

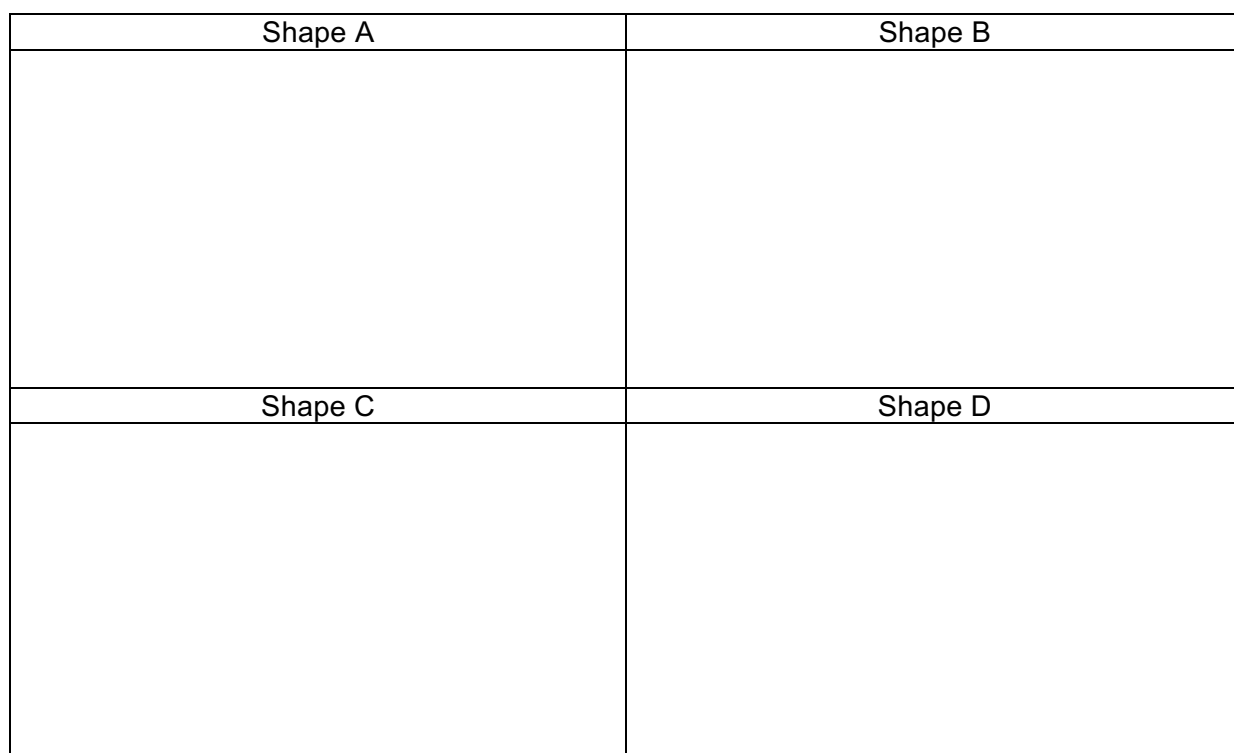
- (a) Match each shape to the top views shown below. Note that two shapes will have the same top view:

1	2
	
3	4
	

The diagrams for each shape have been copied below.

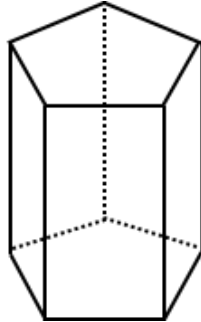


- (b) If the left face is considered to be the front of the shape, draw the front view for each shape below:



Practice OLNA Questions

The diagram below is required to answer questions 1 to 3.



Question 1

How many vertices are there on the three dimensional shape shown above?

- | | | | |
|----------|----|----------|----|
| A | 7 | B | 10 |
| C | 15 | D | 9 |

Question 2

The height of the shape is 8 cm and the area of its base is 12 cm^2 . Given that the volume of a prism is given by the formula, Area (base) $\times$ height, and $1 \text{ cm}^3 = 1 \text{ mL}$, determine how many spoonfuls of honey, each containing 5 mL, could fit into the shape shown above.

- | | | | |
|----------|----|----------|----|
| A | 19 | B | 91 |
| C | 96 | D | 22 |

Question 3

Which two dimensional shape does not appear on the prism above?

- | | | | |
|----------|-----------|----------|---------------|
| A | rectangle | B | parallelogram |
| C | pentagon | D | rhombus |

Graphs and Tables

Two Way Tables

1. The table below shows the sports options chosen by students at the Hightown Middle School

Year Level	Athletics	AFL 9s	Basketball
7	34	48	25
8	19	44	31
9	26	39	22

- (a) How many students chose Athletics?
- (b) How many Year 8s did not choose AFL 9s?
- (c) How many Year 7s were doing a sports option at Hightown?
- (d) How many students playing basketball were not in Year 9?
- (e) On one school day all of the basketball students were taken to a carnival. It was also decided to take all Year 7s, who did not play basketball, as spectators. How many Middle School students remained at school?
2. Ninety-three people worked on the shop floor of the Watson Department Store. The allocation of staff is shown below

	Cosmetics	Mens' Clothing	Ladies' Clothing	Kitchenware	Electronics	TOTAL
Female	13				8	63
Male		7	0	9		
TOTAL	16	13		24		

- (a) Complete the table by filling in the missing information.
- (b) What is the probability that a randomly chosen person works in Kitchenware?
- (c) What fraction of people worked in Cosmetics or Ladies' Clothing?
- (d) What fraction of people who work in Mens' Clothing are female?
- (e) What is the probability that a randomly chosen person is a male and working in Electronics?

3. One hundred members of the Western Australian Cricket Association were surveyed as to which type of cricket was their preference for coming to watch at the ground.

	Tests	One Day	T20	Shield	TOTAL
18 - 34 years old		1	20	0	
35 - 50 years old	4		30	2	41
51 or older	19	0			36
TOTAL	25			13	

- Complete the table by filling in the missing information.
- What is the probability that a randomly chosen member is 51 or older?
- What is the probability that a randomly chosen member prefers Test Cricket?
- What percentage of members prefer T20 Cricket?
- What fraction of members are between 18 and 34 and prefer T20 Cricket?
- What is the probability that a randomly chosen member has not selected One Day Cricket as the favourite game to watch live at the ground?

4.

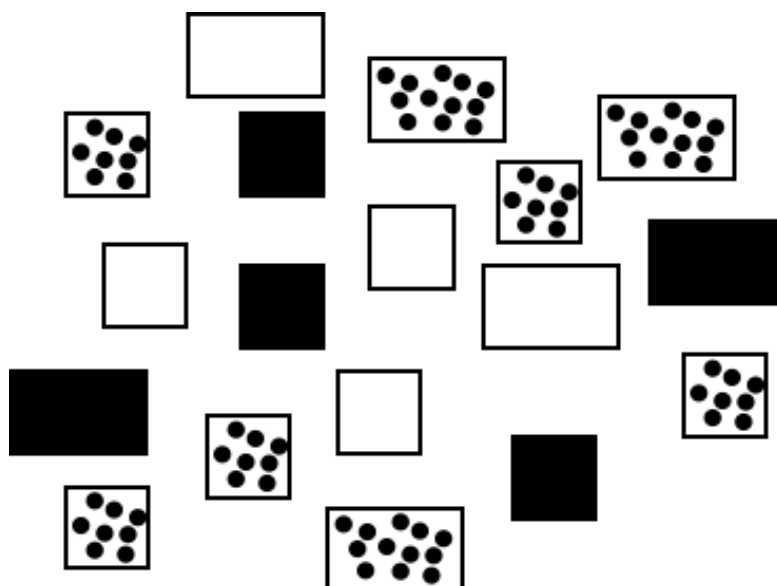


Two spinners are shown above. The table below shows all the possible results for each spinner, and the resulting product of the two spinners

	1	2	3	4
5	5			
6			18	
7				
8				

- Complete the table above.
- How many possible products can occur?
- Which product is more likely, 18 or 24? Why?

5. Consider the shapes shown below:



The diagram above shows squares and rectangles. Some have a black interior, some have dots and others have white interior. The table below summarises the colour and shape.

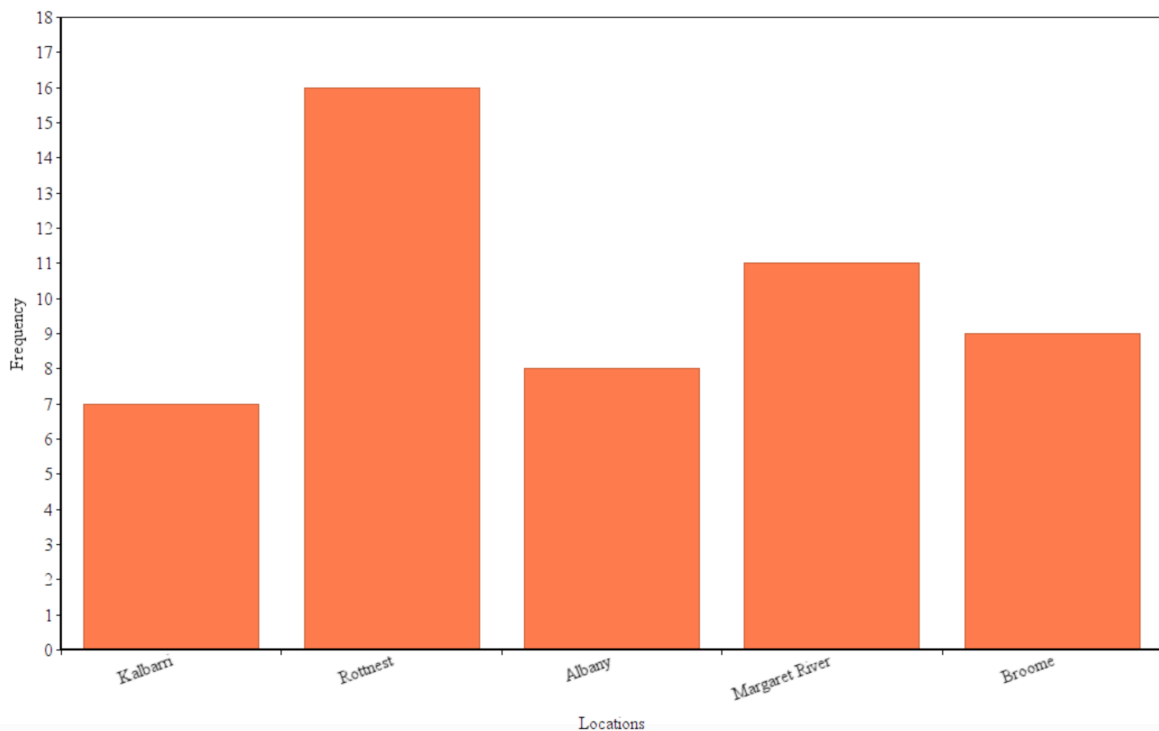
	Squares	Rectangles	Total
Black			
Dots			
White			
Total			

- (a) Complete the table above.
- (b) What is the probability that a randomly chosen shape is:
- white inside and is a square,
 - black inside and is a rectangle,
 - a rectangle,
 - one without dots inside,
 - a square, but not white inside.
- (c) If a black shape was selected, is it more likely to be a square or rectangle?

Graphs

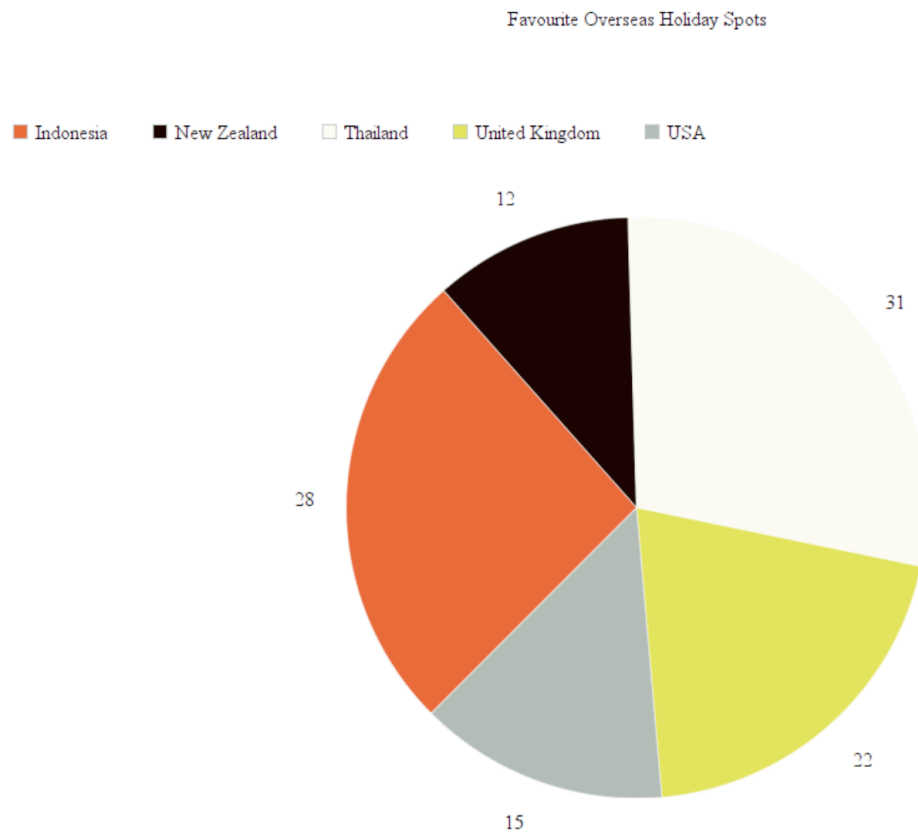
1. A group of Year Ten students from Perth was asked to nominate their favourite holiday destination within Western Australia. The students had to make their choice from five destinations. The graph below displays the results.

Favourite WA Holiday Spots



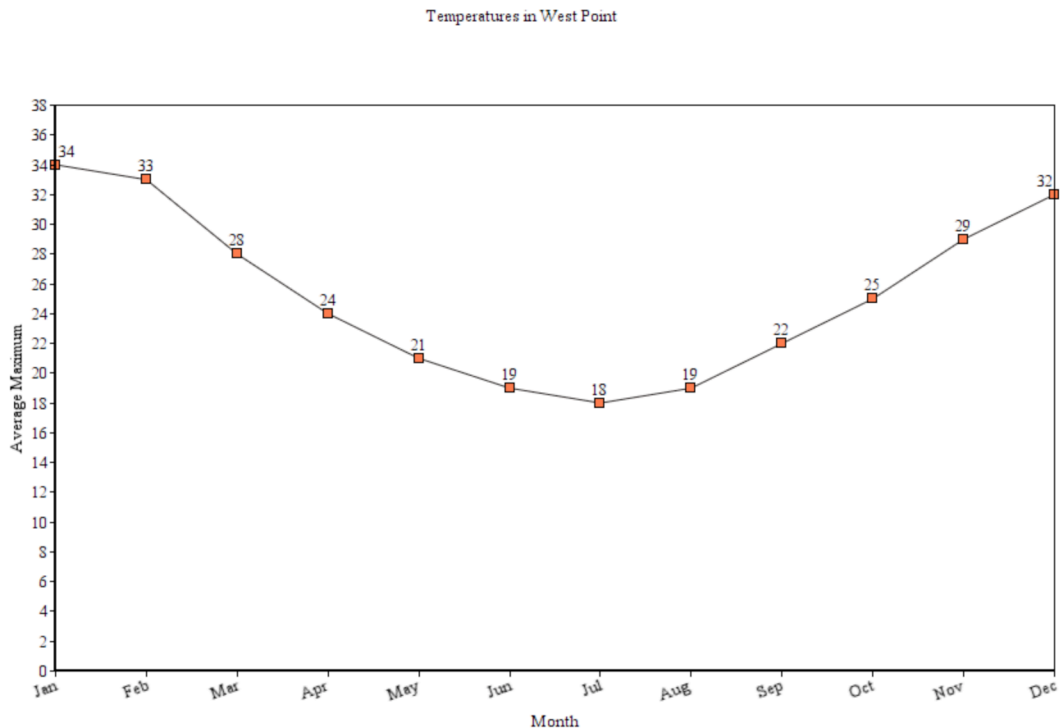
- (a) Which location was the most popular for students in this group?
- (b) How many students did not nominate Rottnest?
- (c) How many students were surveyed?
- (d) How many students preferred Margaret River to Albany?
- (e) What fraction of the class nominated Broome as the preferred holiday spot?

2. Another group of Year Ten students, different from those in Question One, was asked to nominate their favourite overseas holiday destination. The students had to make their choice from five destinations. The graph below displays the results.



- (a) How many students were surveyed?
- (b) How many students stated that New Zealand was their favourite holiday spot?
- (c) Which holiday destination was the most popular choice?
- (d) How many students preferred the United Kingdom to the USA?
- (e) If you divide the number of students by three, then multiply by ten, this determines the size of the angle in the pie graph. What is the angle at the centre for the USA section of the Pie Graph?

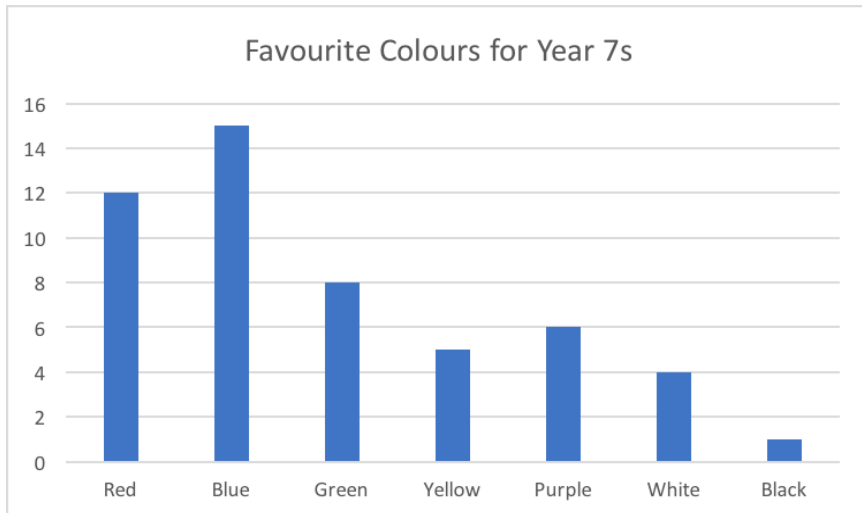
3. West Point is the name for proposed holiday resort on the coast of Western Australia. The temperature graph, which displays the average maximum for each month, is shown below.



- (a) Which month has the highest average maximum?
- (b) Which month has the lowest average maximum?
- (c) What is the difference between the averages for March and June?
- (d) Tourists seeking the sun tend to look for temperatures which exceed 26° . Based on the average temperatures, how many months would be in the peak tourist season for West Point?
- (e) Sid loves travelling along the coast of Western Australia, but does not enjoy the cold. If the monthly maximum for a region is under 22° he avoids travelling in that area. In which months would Sid avoid visiting West Point?

Practice OLNA Questions

Questions 1 to 3 relate to the column graph on favourite colours, shown below.



Question 1

How many Year 7s were surveyed?

- | | | | |
|----------|----|----------|----|
| A | 49 | B | 51 |
| C | 48 | D | 50 |

Question 2

The number of students who liked blue, is greater than those that like yellow by a factor of

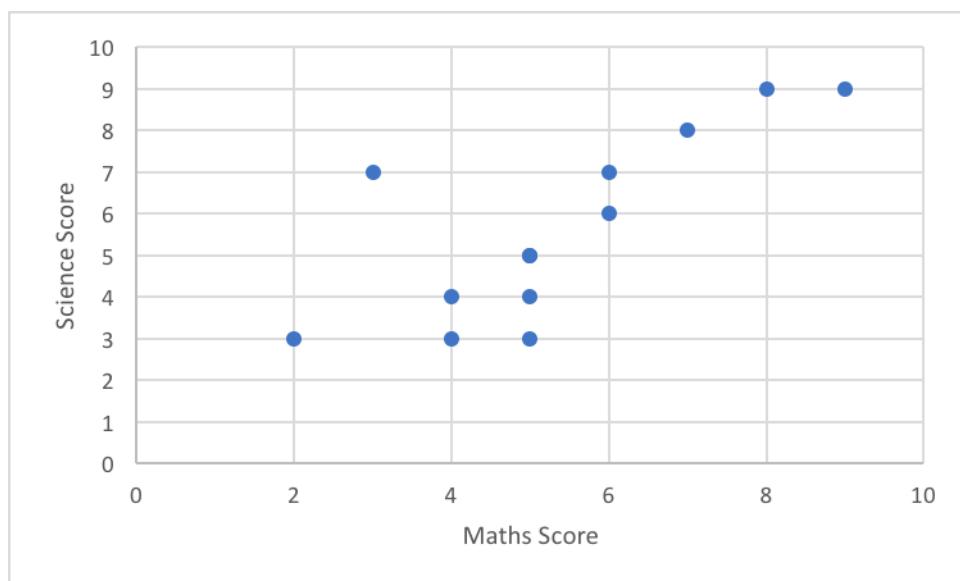
- | | | | |
|----------|----|----------|---|
| A | 10 | B | 2 |
| C | 9 | D | 3 |

Question 3

Which answer below is the best estimate for the percentage of students who prefer green?

- | | | | |
|----------|------|----------|------|
| A | 16 % | B | 10 % |
| C | 8 % | D | 24 % |

Questions 4 to 5 relate to the scattergraph shown below. This graph shows a class of 12 Year 8 students and what they scored in a Maths test, compared to their Science test. Both tests were out of ten.



Question 4

A new student was not included in the graph above, as they only sat the Maths test and were absent for the Science test. If they scored 7 in the Maths test, which of the scores below, would be the best prediction for their Science score?

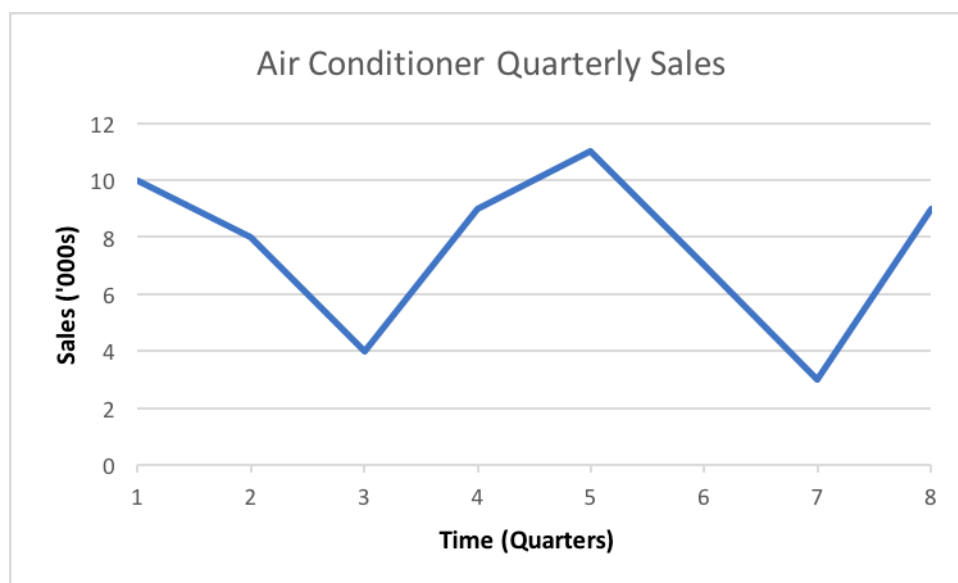
- | | | | |
|----------|---|----------|---|
| A | 3 | B | 5 |
| C | 8 | D | 9 |

Question 5

The Science teacher, Bill, observed that students who scored higher in Science tended to score higher in Maths. A Maths teacher, Stephen, observed that Science scores greater than 6 only occur if a student scored 6 or more in their Maths test.

- | | | | |
|----------|---------------------------------|----------|-------------------------------|
| A | Both observations are true | B | Only Bill's statement is true |
| C | Only Stephens statement is true | D | Both observations are false |

Questions 6 to 8 relate to the line graph shown below. This graph shows the Air Conditioning sales for a company over two years, broken down into eight quarters.



Question 6

If the first quarter represents the sales for the summer season, from December 1 to January 31 of the summer of 2014-2015, state which quarter gave the highest sales.

- | | |
|---------------------------|----------------------|
| A Autumn 2015 | B Spring 2015 |
| C Summer 2015-2016 | D Spring 2016 |

Question 7

What was the total number of sales of air conditioners for the Winter quarters, in 2015 and 2016?

- | | |
|-------------|-----------------|
| A 7 | B 10 000 |
| C 10 | D 7 000 |

Question 8

Based on the graph above, which figure below represents the best estimate for the sales of air conditioners for Autumn 2017?

- | | |
|-------------|-----------------|
| A 7 | B 12 000 |
| C 12 | D 7 000 |



Rounding

Quick Drills

1. Round the numbers in the first column to the nearest value as described in the table below.

Original Number	Nearest Thousand	Nearest Hundred	Nearest Ten
21 590			
21 629			
1 822			
5 237			
12 700			
3 729			
11 986			
14 118			
13 731			
20 088			
6 896			
8 389			
3 335			
11 996			
10 317			

2. Majorie was given \$200 for the outstanding performance of her netball team. She decided to share the money equally with her six team mates. How much money did each person get?
3. The state government gave a grant to the local football club based on their crowd numbers at their home games. Each crowd number was rounded to the nearest hundred and then the calculation of money was based on \$200 per hundred spectators per game. How much money did they receive if their crowds for eight home games were : 934, 1 246, 1 188, 1 009, 1 842, 1 762, 2 389 and 1 849?

Scales

State the temperature shown on the scales below:

Question 1

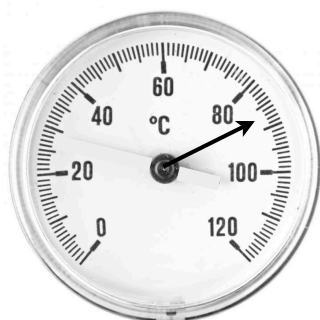
(a)



(b)



(c)



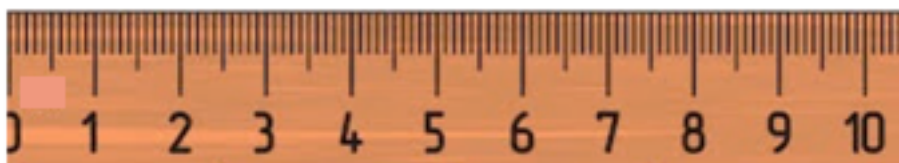
(d)



Question 2

The numbers shown on the ruler below represent centimetres. Show the following measurements on the ruler:

1.5 cm, 56 mm, 9.3 cm, 74 mm



Rates

1. Several Tradespeople compared their charges to determine the costs for a variety of jobs. The costs shown in the table below do not include charges for parts, or equipment.

Call out fee	Charge per hour	Hours worked	Total charge
\$50	\$45	3	
\$60	\$35	4	
\$75	\$65		\$205
\$45	\$55	1.5	
\$90		3	\$270
\$56	\$60	4.5	
	\$45	6	\$380
\$80		2	\$220
\$95	\$80		\$375
\$65		1	\$130
\$70		3	\$235
\$75	\$60		\$405

2. Complete the following table.

Distance	Time	Speed (km/h)
330 km	3 h	110 km/h
100 km	2 h	
55 km	30 min	
	4 h	60 km/h
8 km		40 km/h
24 000 m	0.25 h	
	3.5 h	70 km/h

Practice OLNA Questions

Question 1

The Southern Express can travel at 80 metres per second at its top speed. If it maintains this speed, how far can it go in 3 minutes?

- | | | | |
|----------|--------|----------|---------|
| A | 240 m | B | 14.4 km |
| C | 240 km | D | 24 km |

Question 2

The Perth Scorchers require 84 at a run rate of 10.50 per over. How many overs are remaining?

- | | | | |
|----------|---|----------|---|
| A | 6 | B | 7 |
| C | 8 | D | 9 |

Question 3

The Sydney Sixers know that they can score at 9 runs per over. If their score is 116 and they need to score 175 to win, what will be the outcome of the game, if there are 7 overs remaining?

- | | | | |
|----------|------------|----------|-------------|
| A | Sixers win | B | Sixers lose |
| C | Tied game | | |

Question 4

If a car travels at 90 km/h, how far will it go in 90 minutes?

- | | | | |
|----------|--------|----------|--------|
| A | 120 km | B | 135 km |
| C | 81 km | D | 90 km |

Question 5

In one hour, a factory conveyor belt can produce 7 500 cans of fruit. How many cans will it produce in 20 minutes?

- | | | | |
|----------|-------|----------|-------|
| A | 150 | B | 750 |
| C | 1 500 | D | 2 500 |

Question 6

How much will 10 kg of apples cost if 4 kg cost \$11?

- | | | | |
|----------|---------|----------|------|
| A | \$16.50 | B | \$55 |
| C | \$27.50 | D | \$44 |

Question 7

A car travels at 80 km/h. How long does it take to travel 440 km?

- | | | | |
|----------|---------------------|----------|---------------------|
| A | 5 hours, 30 minutes | B | 5 hours, 40 minutes |
| C | 6 hours | D | 5 hours, 20 minutes |

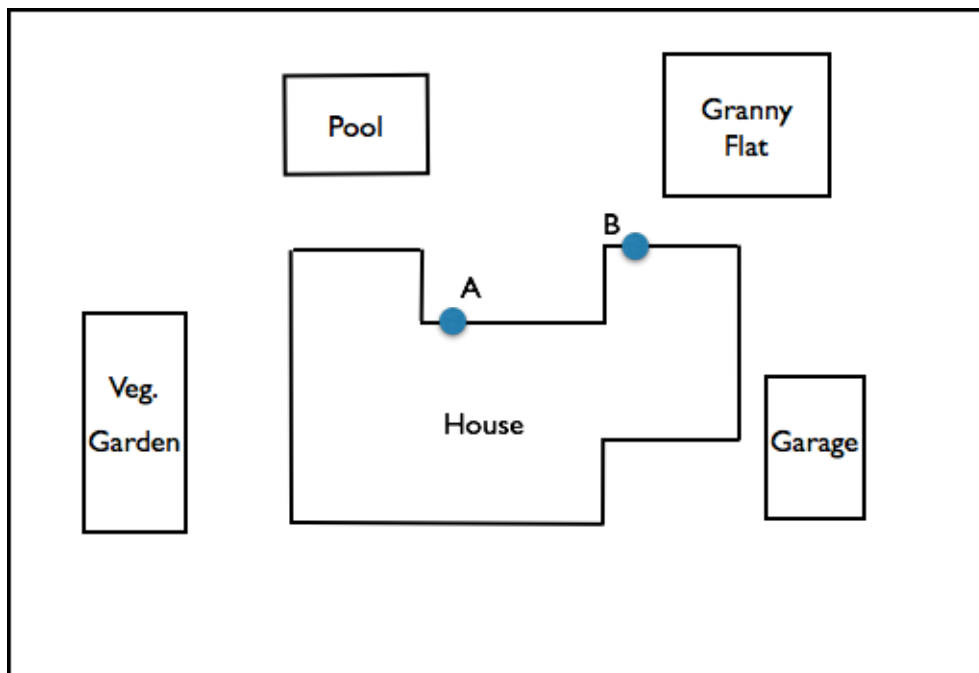
Question 8

What is the cost of 6.5 m of carpet, at \$52 per metre?

- | | | | |
|----------|-------|----------|-------|
| A | \$301 | B | \$315 |
| C | \$338 | D | \$326 |

Maps

The map below shows a scale drawing for a house on a rectangular block.
The scale is 1 : 400



Use the map and scale to answer the following questions.

1. If the house is approximately 24 m wide at its widest point, how wide will the house be, in centimetres, on the map?
2. The house has an exit point at B, which is 2 m to the entry of the Granny Flat. How far will that be, in centimetres, on the map?
3. The Vegetable Garden has a perimeter of 8 cm on the map. What is the actual perimeter, in metres, of this garden?
4. The pool has a length of approximately 2 cm on the map. How far would you swim if you went up and back, ten times?

5. A small truck wants to be able to fit into the garage. Its width is 5.4 m. If the garage is 1.5 cm wide on the map, will the truck be able to fit?

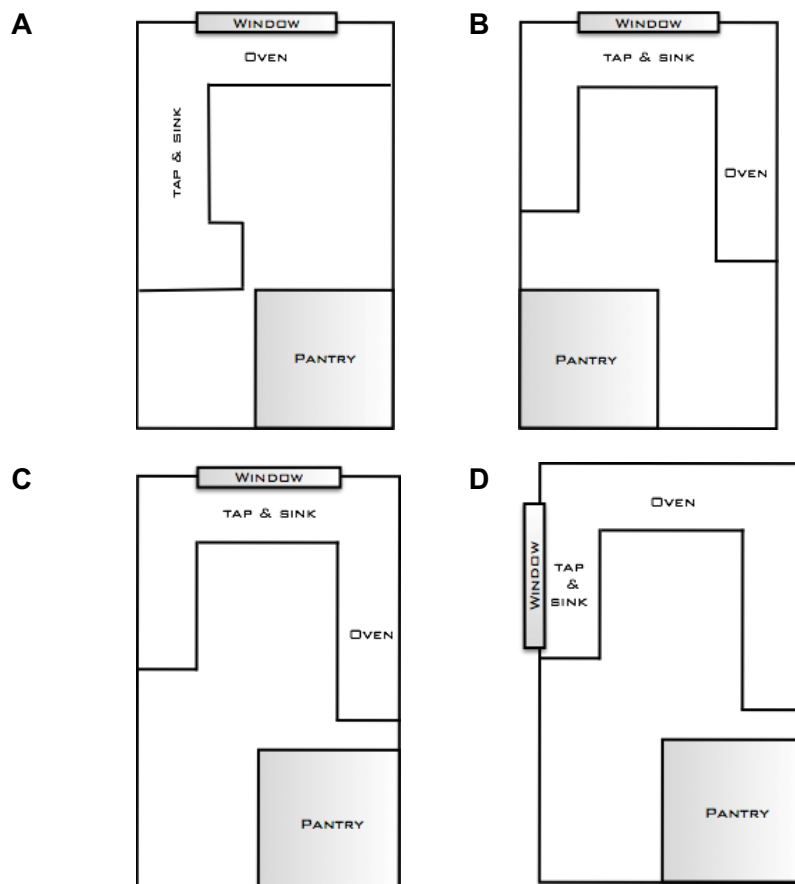
Practice OLNA Questions

The diagram below shows a camera looking into a kitchen from the dining room, in a new home.



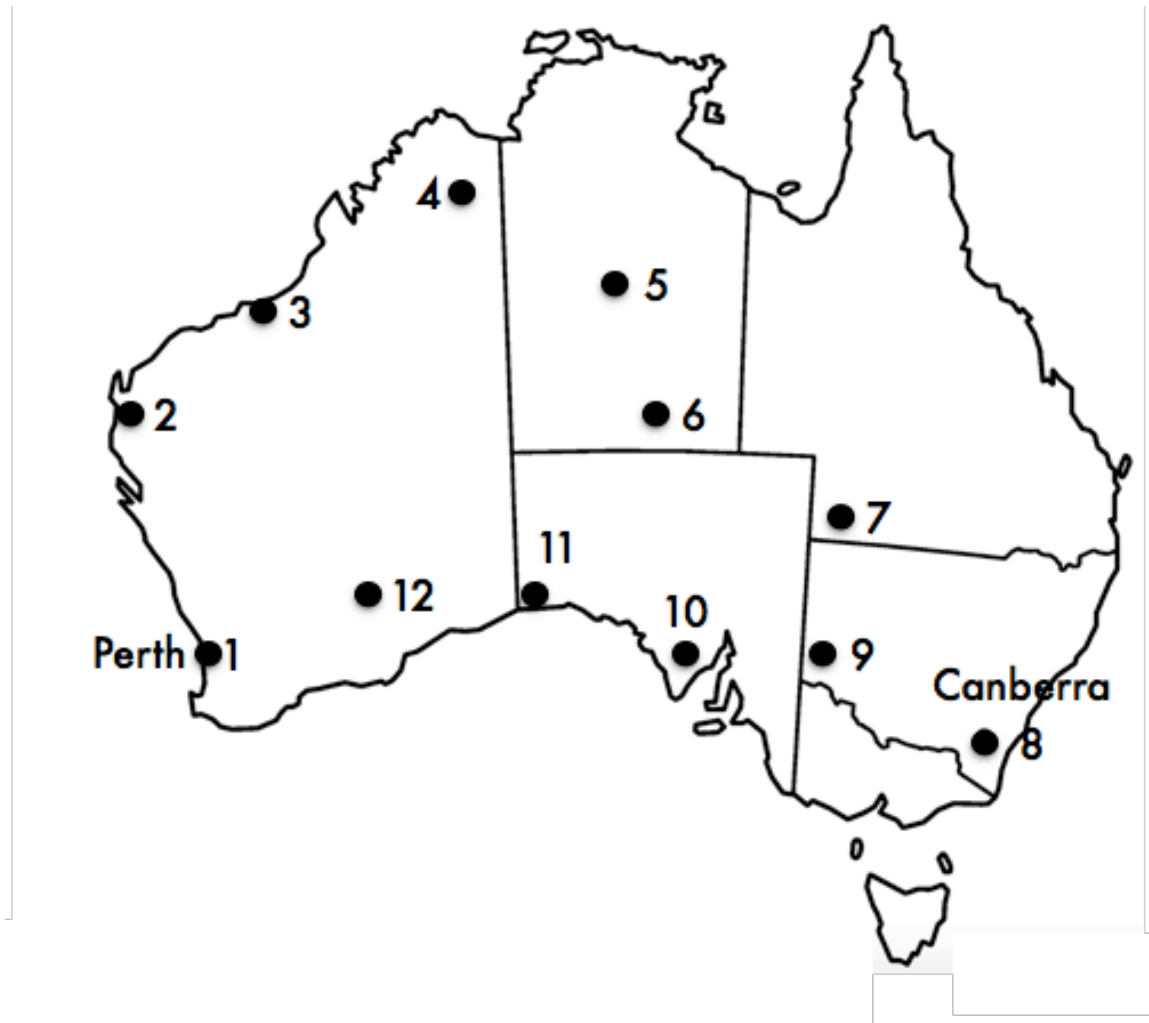
Question 1

Which plan below is the best match for the picture shown above?



Jonty and Debbie recently immigrated to Australia and decided to travel around Australia. The map below shows their journey, starting in Perth and heading as far east as Canberra. Each point represents a stopping point in their journey and the numbers indicate where they started each day of travel. For example, Day 1 started from Perth.

The map shown below shows an approximate scale of 1 cm to 300 km. Use this map to answer questions 2 to 3.



Question 2

How far did Jonty and Debbie travel on Day 1, if the distance from Point 1 to Point 2, on the map, is 3.5 cm

- | | | | |
|----------|----------|----------|-----------|
| A | 1 050 km | B | 10 500 km |
| C | 950 km | D | 1 150 km |

Question 3

On the morning of Day 4, Jonty and Debbie travelled 150 km to enter the Northern Territory. How far would that distance be on the map?

A 50 mm

B 1 cm

C 0.5 cm

D 3 mm

Question 4

On Day 9, Jonty and Debbie covered a distance equivalent to 2 cm on the map. What is their average speed if they took 8 hours to cover this distance.

A 100 km/h

B 80 km/h

C 75 km/h

D 60 km/h

SOLUTIONS

SOLUTIONS : Money

GST

1. (a) \$66 (b) \$25 000 (c) \$198
- 2.

Price of Taxable Item	GST	Total
\$100	\$10	\$110
\$10	\$1	\$11
\$20	\$2	\$22
\$22.50	\$2.25	\$24.75
\$35	\$3.50	\$38.50
\$1 000	\$100	\$1 100
\$54.20	\$5.42	\$59.62
\$145	\$14.50	\$159.50
\$15	\$1.50	\$16.50
\$230.55	\$23.06	\$253.61
\$1 094	\$109.40	\$1 203.40
\$200	\$20	\$220
\$55	\$5.50	\$60.50

Change

Cash Given	Sale Price	Change
\$50	\$24.50	\$25.50
\$100	\$79.80	\$20.20
\$10	\$7.55	\$2.45
\$30	\$23.60	\$6.40
\$85	\$82.05	\$2.95
\$50	\$36.95	\$13.05
\$2	\$1.65	\$0.35
\$40	\$28.95	\$11.05
\$20	\$12.85	\$7.15
\$70	\$58.20	\$11.80
\$160	\$144.35	\$15.65
\$200	\$160.85	\$39.15

Discount

Original Price	Percentage Discount	Discount	New Price
\$50	10%	\$5	\$45
\$240	20%	\$48	\$192
\$300	5%	\$15	\$285
\$80	10%	\$8	\$72
\$500	5%	\$25	\$475
\$50	40%	\$20	\$30
\$688	50%	\$344	\$344
\$350	20%	\$70	\$280
\$600	2%	\$12	\$588
\$360	5%	\$18	\$342
\$85	10%	\$8.50	\$76.50
\$49	25%	\$12.25	\$36.75

Practice OLNA Questions: Money

- | | | | |
|-------------|-------------|-------------|-------------|
| 1 B | 2 A | 3 B | 4 D |
| 5 C | 6 B | 7 D | 8 D |
| 9 A | 10 C | 11 D | 12 C |
| 13 B | 14 D | 15 D | 16 C |
| 17 D | 18 C | | |

SOLUTIONS : Time

Calendars

- | | |
|--|--|
| 1. (a) Monday, Tuesday, Wednesday.
(c) 23
(e) Thursday | (b) $\frac{4}{31}$
(d) Thursday |
|--|--|

2.

Su	M	Tu	W	Th	F	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Differences in Time

In each case below determine the difference in time, in minutes. Time 2 is the later time.

SET ONE

Time 1	Time 2	Difference (Minutes)
7:33 am	7:56 am	23 min.
12:11 pm	12:54 pm	43 min.
1:05 am	1:27 am	22 min.
5:01 am	5:39 am	38 min.
6:11 pm	6:36 pm	25 min.
11:15 pm	11:56 pm	41 min.
10:16 am	10:49 am	33 min.
7:45 pm	7:59 pm	14 min.
3:01 am	3:28 am	27 min.
5:26 pm	6:00 pm	34 min.

SET TWO

Time 1	Time 2	Difference (Minutes)
3:27 am	4:19 am	52 min.
5:16 am	6:08 am	52 min.
10:24 pm	11:36 pm	72 min.
1:19 am	2:09 am	50 min.
3:51 pm	4:23 pm	32 min.
12:16 am	1:23 am	67 min.
3:24 pm	4:01 pm	37 min.
6:09 pm	7:27 pm	78 min.
8:18 am	9:51 am	93 min.
4:02 pm	5:06 pm	64 min.

SET THREE

Time 1	Time 2	Difference (Minutes)
12:33 am	2:56 am	143 min.
12:24 pm	1:37 pm	73 min.
12:44 pm	2:44 pm	120 min.
12:39 pm	1:56 pm	77 min.
12:04 am	1:01 am	57 min.
12:29 pm	2:34 pm	125 min.
12:18 am	3:19 am	181 min.
12:44 am	2:12 am	28 min.
12:07 pm	4:22 pm	255 min.
12:51 am	3:25 am	154 min.

SET FOUR

Time 1	Time 2	Difference (Minutes)
10:14 pm	1:20 am	186 min.
11:26 pm	12:35 am	69 min.
11:15 am	2:19 pm	184 min.
12:18 pm	1:48 pm	98 min.
11:27 am	1:07 pm	100 min.
9:38 am	1:15 pm	217 min.
11:29 am	2:38 pm	189 min.
10:56 pm	12:21 am	85 min.
8:44 am	1:54 pm	310 min.
6:23 pm	1:04 am	401 min.

Practice OLNA Questions: Time

- | | | | |
|----------------------------|---------|---------|---------|
| 1 B | 2 D | 3 A | 4 B |
| 5 D | 6 D | 7 B | 8 C |
| 9 C | 10 C | 11 C | 12 C |
| 13 D | 14 C | 15 B | 16 D |
| 17 2nd, 1st, 4th, 3rd. | | | |
| 18 D | 19 A | 20 B | 21 C |
| 22 A | 23 B | | |

SOLUTIONS : Number

Quick Drills

7	X	11	=	77
5	X	8	=	40
9	X	3	=	27
2	X	12	=	24
6	X	12	=	72
11	X	10	=	110
9	X	4	=	36
3	X	3	=	9
2	X	4	=	8
7	X	3	=	21

3	X	4	=	12
9	X	5	=	45
4	X	6	=	24
12	X	6	=	72
2	X	12	=	24
6	X	12	=	72
9	X	11	=	99
12	X	2	=	24
9	X	6	=	54
10	X	12	=	120

20	÷	10	=	2
2	÷	1000	=	0.002
30	÷	100	=	0.3
9000	÷	1000	=	9
9	÷	100	=	0.09
7	÷	10000	=	0.0007
800	÷	10	=	80
6	÷	10	=	0.6
0	÷	10	=	0
300	÷	1000	=	0.3
300	÷	10000	=	0.03
1	÷	100	=	0.01
1000	÷	10000	=	0.1
90	÷	1000	=	0.09
400	÷	100	=	4
8	÷	1 000	=	0.008
200	÷	10	=	20
400	÷	10	=	40
8 000	÷	10 000	=	0.8
6 000	÷	1 000	=	6
7 000	÷	10 000	=	0.7
0	÷	10 000	=	0
200	÷	1 000	=	0.2
9 000	÷	10	=	900
600	÷	1 000	=	0.6
0	÷	1 000	=	0
40	÷	10 000	=	0.004

8	÷	10	=	0.8
100	÷	10	=	10
600	÷	100	=	6
6	÷	10 000	=	0.0006
5 000	÷	100	=	50
8 000	÷	10 000	=	0.8
500	÷	100	=	5
30	÷	100	=	0.3
80	÷	100	=	0.8
80	÷	10	=	8
4	÷	10	=	0.4
9 000	÷	1 000	=	9
60	÷	1 000	=	0.06
700	÷	1 000	=	0.7
2	÷	10 000	=	0.0002
7	÷	1 000	=	0.007
6	÷	10	=	0.6
300	÷	1 000	=	0.3
0	÷	10	=	0
7	÷	10	=	0.7
400	÷	100	=	4
100	÷	1 000	=	0.1
6 000	÷	10	=	600
10	÷	10	=	1
500	÷	10	=	50
800	÷	10 000	=	0.08
7 000	÷	1 000	=	7

27	+	47	=	71
14	+	29	=	43
3	+	19	=	22
15	+	17	=	32
26	+	36	=	62
8	+	26	=	34
23	+	16	=	39
10	+	44	=	54
28	+	11	=	39
6	+	48	=	54
13	+	36	=	49
17	+	26	=	43
20	+	29	=	49
2	+	12	=	14
21	+	23	=	44

12	+	35	+	34	=	81
14	+	27	+	26	=	67
2	+	32	+	23	=	57
16	+	27	+	31	=	74
11	+	23	+	22	=	56
6	+	39	+	19	=	64
8	+	34	+	38	=	80
8	+	32	+	21	=	61
16	+	28	+	23	=	67
1	+	21	+	26	=	48
9	+	19	+	31	=	59
8	+	39	+	28	=	75
15	+	33	+	19	=	67
3	+	25	+	19	=	47
10	+	34	+	20	=	64

32	-	5	=	27
33	-	3	=	30
25	-	13	=	12
43	-	17	=	26
45	-	22	=	23
48	-	29	=	19
32	-	15	=	17
26	-	23	=	3
20	-	6	=	14
46	-	17	=	29
20	-	4	=	16
35	-	12	=	23
37	-	18	=	19
40	-	23	=	17
42	-	13	=	29

19	+	31	-	24	=	26
13	+	33	-	19	=	27
3	+	39	-	31	=	11
31	-	24	+	28	=	35
30	-	23	+	29	=	36
15	+	34	-	33	=	16
40	-	22	+	21	=	39
44	-	29	-	9	=	6
19	+	21	-	25	=	15
50	-	31	+	27	=	46
45	-	30	-	6	=	9
41	+	33	-	30	=	44
42	-	19	-	16	=	7
20	+	37	-	35	=	22
42	-	27	-	8	=	7

AFL : Scoring

QUESTION ONE

(Missing Numbers)

Koalas 38, Dugites 4, Time : 3 min.
Devils 8, Lizards 9, Time : 10 min.
Quokkas 23, Bulls 4, Time : 31 min.

Camels 17, Emus 10, Time : 26 min.
Crocs 7, Wombats 2, Time : 7 min.
Goannas 18, Echidnas 11, Time : 18 min.

QUESTION TWO

- (a) 9.3, 8.9, 7.15, 6.21 etc (b) 16.2, 15.8, 14.14, 13.20 etc
(c) 20.4, 19.10, 18.16, 17.22 etc (d) 27.5, 26.11, 25.17, 24.23 etc

Recipes

1.

<i>Ingredients</i>	<i>2 people</i>	<i>10 people</i>	<i>1 person</i>	<i>7 people</i>
<i>eggs</i>	1	5	$\frac{1}{2}$	$3\frac{1}{2}$
<i>sugar</i>	1 tablespoon	5 tablespoons	$\frac{1}{2}$ tablespoon	$3\frac{1}{2}$ tablespoons
<i>milk</i>	$\frac{5}{8}$ cup	$2\frac{1}{2}$ cups	$\frac{5}{16}$ cup	$1\frac{31}{42}$ cups
<i>vanilla essence</i>	2 drops	10 drops	1 drop	7 drops
<i>shortcrust pastry</i>	$\frac{1}{2}$ quantity	$2\frac{1}{2}$ quantity	$\frac{1}{2}$ quantity	$1\frac{31}{42}$ quantity
<i>jam</i>	$\frac{1}{2}$ tablespoon	$2\frac{1}{2}$ tablespoons	$\frac{11}{42}$ tablespoon	$1\frac{31}{42}$ tablespoons

2.

<i>Ingredients</i>	<i>2 people</i>	<i>3 people</i>	<i>14 people</i>	<i>9 people</i>
<i>peaches</i>	1	$1\frac{1}{2}$	7	$4\frac{1}{2}$
<i>raspberries</i>	75 mL	112.5 mL	525 mL	337.5 mL
<i>ice cream</i>	2 scoops	3 scoops	14 scoops	9 scoops
<i>sugar</i>	1 tablespoon	$1\frac{1}{2}$ tablespoons	7 tablespoons	$4\frac{1}{2}$ tablespoons
<i>water</i>	50 mL	75 mL	350 mL	225 mL
<i>vanilla</i>	$\frac{1}{2}$ teaspoon	$\frac{11}{42}$ teaspoon	$3\frac{1}{2}$ teaspoons	$\frac{3}{4}$ teaspoons

Triple Treats

- 15. $5 \times 3 \div 1$ OR $5 \div 1 \times 3$
- 6. $12 \div 6 + 4$
- 11. $7 + 10 - 6$
- 17. $5 \times 6 - 13$
- 23. $5 \times 4 + 3$
- 30. $7 \times 3 + 9$
- 5. $2 \times 10 \div 4$
- 21. $9 \div 3 \times 7$
- 42. $(4 + 2) \times 7$
- 8. $10 \div 2 + 3$

Awesome Foursomes

- 40. $10 \div 2 + 5 \times 7$
- 27. $12 \div 4 \times 8 + 3$
- 20. $10 + 4 + 6 \times 1$
- 7. $16 \div 4 + 9 \div 3$
- 34. $(7 - 4) \times 8 + 10$
- 7. $12 \div 6 + 5 \times 1$
- 18. $8 + 5 \times (4 - 2)$
- 30. $(15 - 10) \times 3 \times 2$
- 0. $(24 - 6 \times 4) \times 1$
- 14. $12 \times 5 \div 6 + 4$

Practice OLNA Questions : Number

- | | | | | | | | |
|----------|----------|-----------|----------|-----------|----------|----------|----------|
| 1 | C | 2 | C | 3 | D | 4 | B |
| 5 | B | 6 | B | 7 | D | 8 | D |
| 9 | C | 10 | C | 11 | B | | |

SOLUTIONS : Fractions and Percentages

Quick Drills

SET ONE

20 % of 200 = 40
50 % of 400 = 200
10 % of 1 000 = 100
90 % of 300 = 270
5 % of 600 = 30
60 % of 300 = 180
5 % of 50 = 2.5
35 % of 250 = 87.5
85 % of 450 = 382.5
80 % of 400 = 320
65 % of 300 = 195
75 % of 450 = 337.5
55 % of 500 = 275
10 % of 450 = 45
80 % of 450 = 360

SET TWO

40 out of 200 = 20%
5 out of 500 = 1%
25 out of 500 = 5%
89 out of 100 = 89%
240 out of 1 200 = 20%
30 out of 300 = 10%
45 out of 900 = 5%
75 out of 300 = 25%
225 out of 300 = 75%
200 out of 800 = 25%
10 out of 1 000 = 1%
2.5 out of 500 = 0.5%
40 out of 2 000 = 2%
5 out of 25 = 20%
16 out of 40 = 40%

Practice OLNA Questions : Fractions and Percentages

1	D	2	D	3	A	4	A
5	D	6	C	7	A	8	D

SOLUTIONS : Measurement and Conversions

Conversions

1.

Length (km)	Length (m)
6	6 000
0.9	900
3	3 000
100	100 000
500	500 000
7	7 000
0.2	200
0.6	600
30	30 000
1	1 000
0.4	400
8	8 000
0.7	700
8.1	8 100
0.1	100

Length (cm)	Length (mm)
9	90
6	60
100	1 000
7	70
900	9 000
50	500
700	7 000
8 000	80 000
700	7 000
8 020	80 200
1	10
300	3 000
0.8	8
80	800
30	300

2. (i) 29 cm (ii) 38 000 000 mm (iii) 56 cm
3. (i) 100 276 cm (ii) 666 cm
- (iii) 300 138 cm (iv) 1 127 cm
- (v) 612 600 cm

4.

Capacity (L)	Capacity (mL)
6	6 000
4.2	4 200
8	8 000
300	300 000
0.065	65
0.03	30
40	40 000
23 000	23 000 000
0.04	40
31.277	31 277

Time (hours)	Time (minutes)
4	240
1.5	90
14	840
6 h 20 min	380
7.5	450
3.25	195
0.2	12
2.25	135
20	1 200
6.75	405

Practice OLNA Questions : Measurements and Conversions

1 **B, A, D, C.**

2 **B**

3 **D**

4 **D**

5. **D**

6. **C**

7. **A**

8. **D**

9. **A**

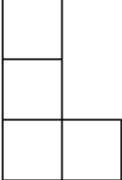
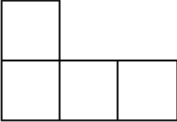
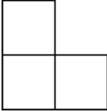
10. **C**

11. **A**

SOLUTIONS : Shapes

1. (a) A matches 2 or 4, B 3, C 1, D 2 or 4.

(b)

Shape A	Shape B
	
Shape C	Shape D
	

Practice OLNA Questions : Shapes

- 1 B 2 A 3 D

SOLUTIONS : Graphs and Tables

Two Way Tables

1. (a) 79 (b) 50 (c) 107
(d) 56 (e) 128

2. (a)

	Cosmetics	Mens' Clothing	Ladies' Clothing	Kitchenware	Electronics	TOTAL
Female	13	6	21	15	8	63
Male	3	7	0	9	11	30
TOTAL	16	13	21	24	19	93

- (b) $\frac{24}{93}$ (c) $\frac{37}{93}$ (d) $\frac{6}{13}$
(e) $\frac{11}{93}$

3. (a)

	Tests	One Day	T20	Shield	TOTAL
18 - 34 years old	2	1	20	0	23
35 - 50 years old	4	5	30	2	41
51 or older	19	0	6	11	36
TOTAL	25	6	56	13	100

(b) $\frac{36}{100}$

(c) $\frac{25}{100}$

(d) $\frac{56}{100}$

(e) $\frac{20}{100}$

(f) $\frac{94}{100}$

4. (a)

	1	2	3	4
5	5	10	15	20
6	6	12	18	24
7	7	14	21	28
8	8	16	24	32

(b) There are 16 products, one of which is repeated.

(c) 24 is more likely as there are two ways this can occur (3×8 , 4×6), whereas 18 can only occur one way.

5. (a)

	Squares	Rectangles	Total
Black	3	2	5
Dots	5	3	8
White	3	2	5
Total	11	7	18

(b) (i) $\frac{3}{18}$
(iv) $\frac{10}{18}$

(ii) $\frac{2}{18}$
(v) $\frac{8}{18}$

(iii) $\frac{7}{18}$

(c) A square

Graphs

1. (a) Rottneest

(b) 35

(c) 51

(d) 3 (11 to 8)

(e) $\frac{9}{51}$

2. (a) 108

(b) 12

(c) Thailand

(d) 7 (22 to 15)

(e) 50°

3. (a) January

(b) July

(c) 9°

(d) 5

(e) May, June, July and August

Practice OLNA Questions : Graphs and Tables

- 1 B 2 D 3 A 4 C
- 5 B 6 C 7 D 8 D

SOLUTIONS : Rounding

Quick Drills

1. Round the numbers in the first column to the nearest value as described in the table below.

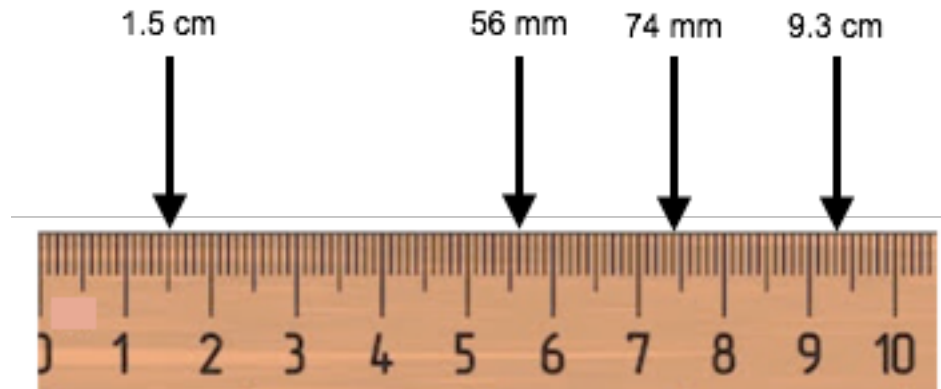
Original Number	Nearest Thousand	Nearest Hundred	Nearest Ten
21 590	22 000	21 600	21 590
21 629	22 000	21 600	21 630
1 822	2 000	1 800	1 820
5 237	5 000	5 200	5 240
12 700	13 000	12 700	12 700
3 729	4 000	3 700	3 730
11 986	12 000	12 000	11 990
14 118	14 000	14 100	14 120
13 731	14 000	13 700	13 730
20 088	20 000	20 100	20 090
6 896	7 000	6 900	6 900
8 389	8 000	8 400	8 390
3 335	3 000	3 300	3 340
11 996	12 000	12 000	12 000
10 317	10 000	10 000	10 320

2. \$33.33 each.
OR to nearest 5 c, it would be \$33.30 each
(We can't round to \$33.35 as there would be insufficient money for everyone.)
3. \$24 200

SOLUTIONS : Scales

1. (a) 50° (b) 69° (c) 87° (d) 8°

2.



SOLUTIONS : Rates

1. Several Tradespeople compared their charges to determine the costs for a variety of jobs. The costs shown in the table below do not include charges for parts, or equipment.

Call out fee	Charge per hour	Hours worked	Total charge
\$50	\$45	3	\$185
\$60	\$35	4	\$200
\$75	\$65	2	\$205
\$45	\$55	1.5	\$127.50
\$90	\$60	3	\$270
\$56	\$60	4.5	\$326
\$110	\$45	6	\$380
\$80	\$70	2	\$220
\$95	\$80	3.5	\$375
\$65	\$65	1	\$130
\$70	\$55	3	\$235
\$75	\$60	5.5	\$405

2. Complete the following table.

Distance	Time	Speed (km/h)
330 km	3 h	110 km/h
100 km	2 h	50 km/h
55 km	30 min.	110 km/h
240 km	4 h	60 km/h
8 km	12 min.	40 km/h
24 000 m	0.25 h	96 km/h
245 km	3.5 h	70 km/h

Practice OLNA Questions : Rates

- | | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | B | 2 | C | 3 | A | 4 | B |
| 5 | D | 6 | C | 7 | A | 8 | C |

SOLUTIONS : Maps

- | | | | |
|----|-------------------------------|----|-------|
| 1. | 6 cm | 2. | 2 cm |
| 3. | 32 m | 4. | 160 m |
| 5. | Yes - the garage is 6 m wide. | | |

Practice OLNA Questions : Maps

- | | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | C | 2 | A | 3 | C | 4 | C |
|---|---|---|---|---|---|---|---|